CONTINUOUS AND SMOOTH PLURISUBHARMONIC EXHAUSTIONS: THREEFOLD COUNTEREXAMPLES AND SURFACE REGULARIZATION

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Abstract. We construct Kähler and non-Kähler threefolds with locally Lipschitz psh exhaustions but no C^1 psh exhaustion; the Kähler example is quasi-projective. For line bundles over compact complex manifolds, the equivalence between semipositive metrics and psh exhaustions of the dual total space preserves C^r, local Hölder, and real analytic regularity. We also prove smooth regularization for divisor complements in compact Kähler surfaces and a holomorphic convexity criterion for weakly complete spaces fibered by curves over affine space.

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Date: November 3, 2026.

2020 *Mathematics Subject Classification.* Primary 32F17; Secondary 32Q15, 32L05, 32U40.

Key words and phrases. Plurisubharmonic exhaustion, Kähler manifold, semipositive line bundle, ruled surface, Hironaka modification, Ueda theory, class VII surface, boundary divisor, positive current.

1. Introduction and main results

A continuous plurisubharmonic exhaustion need not admit a smooth replacement, even on a quasi-projective manifold. We construct threefold examples and prove affirmative results for compactifiable surfaces and spaces fibered by curves. Throughout, psh means plurisubharmonic and smooth means C^∞.

Theorem 1.1. *There exists a connected smooth quasi-projective Kähler threefold M admitting a locally Lipschitz psh exhaustion but no C^1 psh exhaustion.*

The construction uses Grauert's ruled surfaces [Gra62, §4, 8(d)], including Koike's modified example [Koi21, Example 4.1]. For their distinguished nef and big line bundles, we construct locally Lipschitz semipositive metrics, smooth with strictly positive curvature off the negative section, and prove that no semipositive C^1 metric exists near that section. The proof strengthens the smooth first-order obstruction [Koi21, Theorem 1.4] by testing C^1 weights on holomorphic graphs. The regularity criterion (Theorem 3.4) then gives, for every line bundle A over a compact complex manifold,

$$\mathrm{Tot}(A^{-1}) \text{ has a psh exhaustion of class } \mathcal{R}$$
$$\Longleftrightarrow$$
$$A \text{ has a semipositive metric of class } \mathcal{R},$$

Here $\mathcal{R}$ is any of the C^r, $C^{k,\alpha}_{\mathrm{loc}}$, or C^ω classes in (3.1). In particular, the obstruction applies to every C^1 psh exhaustion, without an invariance assumption on the exhaustion.

Theorem 1.2. *Let M be the threefold in Theorem 1.1. For every $x \in M$ there is a smooth non-Kähler threefold N and a proper holomorphic modification $\mu : N \to M$ such that*

$$N \setminus \mu^{-1}(x) \simeq M \setminus \{x\}.$$

The threefold N admits a locally Lipschitz psh exhaustion, but no C^1 psh exhaustion.

Hironaka's opposite-order blowups produce the non-Kähler configuration over x. Proposition 4.1 transfers the exhaustion properties through this compactly supported modification.

Theorem 1.3. *Let S be a smooth compact connected Kähler surface and $D \neq 0$ an effective divisor. If $S \setminus \operatorname{supp} D$ admits a continuous psh exhaustion, it admits a smooth one. Consequently, the same implication holds for every smooth complex quasi-projective surface.*

The proof averages positive dd^c-closed currents over exhaustion levels to obtain a current with zero Lelong numbers. The current theorem of Dinh–Nguyên–Sibony [DNS22] and Hodge index then give the following alternative: either $S \setminus \operatorname{supp} D$ is a modification of a Stein space, or a divisor $A = \sum a_i D_i$, $a_i \in \mathbb{Z}_{>0}$, satisfies

$$\operatorname{supp} A = \operatorname{supp} D, \qquad A^2 = 0, \qquad \mathcal{O}_S(A) \text{ is unitary flat near } D.$$

Its canonical section yields a pluriharmonic logarithmic collar. Section 5 also gives separate numerical and neighborhood criteria for existence of an exhaustion.

Theorem 1.4. *Let X be a reduced complex space of pure dimension $n \geq 2$ with a continuous real-valued psh exhaustion. If a holomorphic map*

$$F : X \longrightarrow \mathbb{C}^{n-1}$$

has every nonempty fiber of dimension at most one, then X is holomorphically convex and admits a smooth psh exhaustion.

The fiber bound includes exceptional fibers. Analytic fiber cycles and Bishop compactness establish the stability needed for Vîjîitu's convexity criteria [Vîj22b]. For irreducible surfaces, convexity already follows from [Vîj22a, Theorem 1.1].

Regularization here means existence of a smooth exhaustion. The problem for arbitrary noncompact complex surfaces remains open. Among smooth quasi-projective manifolds, Theorems 1.1 and 1.3 place the least possible counterexample dimension at three.

Conventions. We use

$$d^c = \frac{i}{4\pi}(\bar\partial - \partial), \qquad dd^c = \frac{i}{2\pi}\partial\bar\partial, \qquad \|e\|_h^2 = e^{-\varphi}, \qquad \Theta_h(A) = dd^c\varphi.$$

For local frames $e_j = g_{ij}e_i$,

$$\varphi_j = \varphi_i - \log|g_{ij}|^2. \tag{1.1}$$

A continuous Hermitian metric is semipositive when its weights are psh; below C^2 the curvature inequality is understood in the sense of currents. All regularity classes are local in real coordinates; $C^{0,1}_{\mathrm{loc}}$ means locally Lipschitz and C^ω means real analytic. For C^2 functions, write $\mathcal{L}_f(\xi) = \sum f_{z_j\bar z_k}\xi_j\overline{\xi_k}$. For the canonical section s_D,

$$dd^c \log\|s_D\|_h^2 = [D] - \Theta_h(\mathcal{O}(D)). \tag{1.2}$$

An exhaustion is a continuous real-valued function with compact sublevel sets. Continuous or smooth weak completeness means existence of a psh exhaustion of that regularity. On complex spaces, smooth functions are local restrictions of smooth ambient functions. We write $\mathbf{P}(E) = \mathbf{P}_{\mathrm{lines}}(E)$ for the projectivization of lines, with tautological bundle $\mathcal{O}_{\mathbf{P}(E)}(-1)$.

2. The ruled surface and its metrics

2.1. **Grauert's construction and its line bundle.** Let B be a smooth connected projective curve of genus $g \geq 2$, let F be a line bundle of degree $d > 0$, and choose $0 \neq \xi \in H^1(B, F)$. Let

$$0 \longrightarrow F \longrightarrow E \longrightarrow \mathcal{O}_B \longrightarrow 0 \tag{2.1}$$

represent ξ. Set

$$S = \mathbf{P}_{\mathrm{lines}}(E), \qquad \pi : S \to B, \qquad Y = \mathbf{P}(F), \qquad L = \mathcal{O}_S(Y) \otimes \pi^*F. \tag{2.2}$$

GAGA algebraizes E, so S is a smooth projective surface. For example, $F = K_B$ always supplies such data, since

$$\deg K_B = 2g - 2 > 0, \qquad H^1(B, K_B) \simeq H^0(B, \mathcal{O}_B)^* \simeq \mathbb{C}.$$

For the original degree-one construction, take $g = 2$ and $F = \mathcal{O}_B(p)$, $p \in B$. A nonconstant meromorphic function with at most one simple pole would define a degree-one map $B \to \mathbf{P}^1$; thus $h^0(B, \mathcal{O}_B(p)) = 1$, and Riemann–Roch gives

$$h^0(B, \mathcal{O}_B(p)) - h^1(B, \mathcal{O}_B(p)) = 1 + 1 - 2 = 0.$$

The choices $F = \mathcal{O}_B(p)$ and $F = K_B$ therefore recover the original and modified Grauert examples, respectively [Gra62][Koi21, Example 4.1].

Choose local frames f_i of F on $U_i \subset B$ and lifts v_i of $1 \in \mathcal{O}_B$, with

$$f_j = g_{ij}f_i, \qquad v_j = v_i + c_{ij}f_i.$$

On $S \setminus Y$, write a line as $[t_if_i + v_i]$. Near Y, write it as $[f_i + w_iv_i]$, so that $w_i = 1/t_i$. Comparing coefficients on overlaps gives

$$t_i = g_{ij}t_j + c_{ij}, \qquad w_i = \frac{w_j}{g_{ij} + c_{ij}w_j}, \qquad w_j = \frac{g_{ij}w_i}{1 - c_{ij}w_i}. \tag{2.3}$$

The quotient map $E \to \mathcal{O}_B$ induces a section of $\mathcal{O}_S(1)$ whose zero divisor is Y. Consequently

$$\begin{aligned} \mathcal{O}_S(Y) \simeq \mathcal{O}_S(1), \qquad & N_{Y/S} \simeq \mathrm{Hom}(F, E/F) \simeq F^{-1}, \\ Y^2 = -d, \qquad & L|_Y \simeq F^{-1} \otimes F \simeq \mathcal{O}_Y. \end{aligned} \tag{2.4}$$

Let d_i be the frame of $\mathcal{O}_S(Y)$ defined by $s_Y = w_i d_i$, and put $\ell_i = d_i \otimes \pi^* f_i$. Then

$$d_j = \frac{1 - c_{ij} w_i}{g_{ij}} d_i, \qquad \ell_j = (1 - c_{ij} w_i) \ell_i. \tag{2.5}$$

In particular, the frames $\ell_i|_Y$ agree and give the trivialization in (2.4).

Proposition 2.1. *The line bundle L is nef and big, with*

$$L^2 = d, \qquad L \cdot Y = 0, \qquad \mathrm{Null}(L) = Y.$$

The bundle $A = L^{\otimes 2} \otimes \mathcal{O}_S(-Y)$ is ample. Every section of every positive tensor power of L vanishes along Y; in particular, L is not semiample.

Proof. Let P be a fiber of π. The intersection identities are

$$Y^2 = -d, \quad Y \cdot \pi^* F = d, \quad (\pi^* F)^2 = 0, \quad Y \cdot P = 1, \quad \pi^* F \cdot P = 0.$$

Thus $L^2 = d$, $L \cdot Y = 0$, and $L \cdot P = 1$. For an irreducible horizontal curve $D \neq Y$, let $n = \deg(\pi|_{\widetilde{D}}) > 0$, where $\widetilde{D}$ is its normalization. Then

$$L \cdot D = Y \cdot D + nd > 0.$$

This proves nefness and the null-locus assertion. Moreover

$$\begin{gathered} A = \mathcal{O}_S(Y) \otimes \pi^*(F^{\otimes 2}), \qquad A^2 = 3d, \\ A \cdot Y = d, \qquad A \cdot P = 1, \qquad A \cdot D = Y \cdot D + 2nd > 0. \end{gathered}$$

The Nakai–Moishezon criterion makes A ample. Multiplication by s_Y^k and the Hilbert polynomial give

$$H^0(S, A^{\otimes k}) \hookrightarrow H^0(S, L^{\otimes 2k}), \qquad h^0(S, L^{\otimes 2k}) \geq \frac{3d}{2} k^2 + O(k),$$

so L is big.

We first show that $S \setminus Y$ contains no compact curve. If D were such a curve, its normalization would give a finite surjective map $p : \widetilde{D} \to B$ of degree $n > 0$: a compact curve cannot lie in an affine fiber. Away from the branch values, put

$$a_i(b) = \frac{1}{n} \sum_{x \in p^{-1}(b)} t_i(x).$$

The sum is holomorphic and single-valued, and is locally bounded at branch values by compactness of the inverse image of a small closed disc. It extends holomorphically across them. Equation (2.3) implies

$$a_i = g_{ij} a_j + c_{ij}, \qquad c_{ij} f_i = a_i f_i - a_j f_j.$$

This makes the cocycle representing ξ a coboundary, a contradiction.

Now let $\sigma \in H^0(S, L^{\otimes m})$, $m > 0$. Its restriction to Y is constant. If that constant were nonzero, the zero divisor of σ would be disjoint from Y and would be nonempty, since

$$L^{\otimes m}|_P \simeq \mathcal{O}_{\mathbf{P}^1}(m), \qquad \deg \mathrm{div}(\sigma|_P) = m > 0.$$

It would be a compact curve in $S \setminus Y$, which is impossible. Hence $\sigma|_Y = 0$ for every $m > 0$. $\square$

2.2. The first normal derivative and the C^1 obstruction.

Lemma 2.2 (First normal derivative)**.** *Let $D \subset \mathbb{C}$ be a disc and let φ be a real-valued C^1 psh function on a neighborhood of $D \times \{0\}$ in $\mathbb{C}^2$. If $\varphi(z,0)$ is constant, then $b(z) = \partial_w \varphi(z,0)$ is holomorphic.*

Proof. Subtract the constant and fix a subdisc $D_0 \Subset D$. Put $K = \overline{D}_0$ and choose $\delta > 0$ such that $K \times \{|w| \le \delta\}$ lies in the domain of φ. For $z \in K$ and $|w| \le \delta$, the fundamental theorem of calculus gives

$$\begin{aligned} \varphi(z,w) &= 2\,\mathrm{Re}\left(w \int_0^1 \partial_w \varphi(z,sw)\, ds \right) \\ &= 2\,\mathrm{Re}(b(z)w) + R(z,w), \qquad |R(z,w)| \le 2|w|\omega_K(|w|), \end{aligned} \tag{2.6}$$

where

$$\omega_K(\varepsilon) = \sup_{\substack{z \in K \\ |v| \le \varepsilon}} |\partial_w \varphi(z,v) - \partial_w \varphi(z,0)| \longrightarrow 0.$$

For a holomorphic function f on D, put $A = \max\{1, \sup_K |f|\}$. If $0 < t < \delta/A$, the graph restrictions

$$u_t^{\pm}(z) = \frac{\varphi(z, \pm t f(z))}{t}$$

are subharmonic on D_0, and (2.6) gives

$$\sup_K |u_t^{\pm} \mp 2\,\mathrm{Re}(bf)| \le 2A\omega_K(tA) \longrightarrow 0.$$

Passing to the limit in the submean inequalities shows that both $\mathrm{Re}(bf)$ and its negative are subharmonic; hence $\mathrm{Re}(bf)$ is harmonic on D_0. Since $D_0 \Subset D$ was arbitrary, this holds throughout D. Taking $f = 1, i$ shows that $\mathrm{Re}\, b$ and $\mathrm{Im}\, b$ are harmonic; in particular, $b \in C^\infty(D)$ and $\Delta b = 0$. Taking $f = z, iz$ also makes zb harmonic. We may therefore compute

$$0 = \Delta(zb) = z\Delta b + 4\partial_{\bar{z}} b.$$

Thus $\partial_{\bar{z}} b = 0$. □

Proposition 2.3. *No neighborhood of Y carries a C^1 Hermitian metric on L with psh weights. The same holds for $L^{\otimes m}$, $m \ge 1$.*

Proof. Suppose the weights φ_i in the frames ℓ_i are C^1 and psh near Y. Equation (2.5) gives

$$\varphi_j(z, w_j) = \varphi_i(z, w_i) - \log|1 - c_{ij} w_i|^2. \tag{2.7}$$

The restrictions $\varphi_i|_Y$ glue to a subharmonic function on the compact connected curve Y, hence to a constant. Rescale the metric so that $\varphi_i(z,0) = 0$. By Lemma 2.2, the functions

$$b_i(z) = \partial_{w_i} \varphi_i(z,0)$$

are holomorphic. Differentiate (2.7) once at $w_i = 0$, using (2.3):

$$g_{ij} b_j = b_i + c_{ij}, \qquad v_j - b_j f_j = v_i + (c_{ij} - g_{ij} b_j) f_i = v_i - b_i f_i.$$

Therefore

$$s_i : \mathcal{O}_B|_{U_i} \longrightarrow E|_{U_i}, \qquad a \longmapsto a(v_i - b_i f_i)$$

glue to a holomorphic right inverse of $E \to \mathcal{O}_B$, contradicting $\xi \neq 0$. A C^1 semipositive metric on $L^{\otimes m}$ would give one on L by dividing its weights by m. □

The transition calculation identifies the first obstruction as

$$[\{-c_{ij}\,dw_i\}] \in H^1(Y, N_{Y/S}^{-1}) \simeq H^1(B, F), \qquad [\{-c_{ij}\,dw_i\}] = -\xi.$$

This is the obstruction appearing in [Koi21, Theorem 1.4]; Lemma 2.2 establishes its consequence for C^1 weights.

2.3. **Cusp correction and a global metric.**

Lemma 2.4 (Cusp correction)**.** *Let D be a compact smooth connected curve in a complex surface X, with $D^2 < 0$, and let H be a line bundle near D whose restriction admits a smooth flat metric. There are smooth metrics h_0 on H and k on $\mathcal{O}_X(D)$, a neighborhood U of D, and $\Lambda > 0$ such that*

$$h_{\mathrm{loc}} = h_0 e^{-\Lambda s}, \qquad s = \|s_D\|_k,$$

has locally Lipschitz psh weights on U and smooth strictly psh weights on $U \setminus D$.

Proof. Extend the flat metric on $H|_D$ smoothly and write $\theta = \Theta_{h_0}(H)$; then $\theta|_{TD} = 0$. Since $\deg N_{D/X} = D^2 < 0$, choose a metric on $N_{D/X}$ with strictly negative curvature and extend it to a smooth metric k on $\mathcal{O}_X(D)$. Indeed, if γ is an initial curvature and ω_D is an area form of integral one, the equation

$$dd^c v = D^2\omega_D - \gamma, \qquad \int_D (D^2\omega_D - \gamma) = 0$$

is solvable on the compact curve. Multiplication by e^{-v} gives the prescribed negative curvature. Smooth extensions follow by extending the logarithm of the ratio of two metrics. Put

$$\eta = -\Theta_k(\mathcal{O}_X(D)), \qquad q = \|s_D\|_k^2, \qquad s = q^{1/2}.$$

In coordinates $D = \{w = 0\}$, with $s_D = wd$ and $\|d\|_k^2 = e^\beta$, one has

$$q = |w|^2 e^\beta, \qquad s = |w| e^{\beta/2}, \qquad \eta = dd^c\beta, \qquad \eta|_{TD} > 0. \tag{2.8}$$

Shrink finitely many relatively compact charts covering D until $|w\beta_w| \le 1/2$. Define smooth vector fields

$$H_1 = \partial_z - \frac{w\beta_z}{1 + w\beta_w}\partial_w, \qquad H_2 = \partial_w.$$

They form a frame, with $H_1|_D = \partial_z$, and on $w \neq 0$

$$\partial \log q(H_1) = 0, \qquad \partial \log q(H_2) = \frac{1}{w} + \beta_w.$$

The tensor identity

$$dd^c s = \frac{s}{2}\eta + \frac{s}{4}\frac{i}{2\pi}\partial \log q \wedge \bar\partial \log q \tag{2.9}$$

therefore gives, in the frame (H_1, H_2),

$$\begin{aligned} (dd^c s)_{1\bar 1} &= \tfrac{s}{2}\eta_{1\bar 1}, & (dd^c s)_{1\bar 2} &= \tfrac{s}{2}\eta_{1\bar 2}, \\ (dd^c s)_{2\bar 2} &= \tfrac{s}{2}\eta_{2\bar 2} + \tfrac{s}{4}|w^{-1} + \beta_w|^2. \end{aligned}$$

These coefficients are tensor evaluations in a smooth frame, with the common factor $i/(2\pi)$ omitted. Choose uniform constants $\mu > 0$, $A \ge 1$ with

$$\eta_{1\bar 1} \ge \mu, \qquad |\eta_{1\bar 2}| + |\eta_{2\bar 2}| \le A, \qquad A^{-1} \le e^\beta \le A.$$

Since $|1 + w\beta_w| \ge 1/2$,

$$(dd^c s)_{2\bar 2} \ge -\frac{A}{2}s + \frac{s}{16|w|^2} \ge -\frac{A}{2}s + \frac{1}{16As} \ge \frac{1}{32As}$$

after imposing $s^2 \le 1/(16A^2)$. Smoothness and $\theta_{1\bar1}|_D = 0$ also give $|\theta_{1\bar1}| \le C|w| \le CA^{1/2}s$. Consequently there are $a, b, K > 0$, independent of the chart, such that

$$\begin{aligned} (dd^c s)_{1\bar1} \ge as, \qquad (dd^c s)_{2\bar2} \ge b/s, \quad |(dd^c s)_{1\bar2}| \le Ks, \\ \theta_{1\bar1} \ge -Ks, \qquad \theta_{2\bar2} \ge -K, \qquad |\theta_{1\bar2}| \le K. \end{aligned} \tag{2.10}$$

First fix Λ so that

$$\Lambda a - K > 0, \qquad (\Lambda a - K)\frac{\Lambda b}{2} > (K+1)^2. \tag{2.11}$$

Then shrink U until $Ks \le \Lambda b/2$ and $\Lambda K s \le 1$. For $T = \theta + \Lambda dd^c s$,

$$T_{1\bar1} \ge (\Lambda a - K)s > 0, \qquad T_{2\bar2} \ge \frac{\Lambda b}{2s} > 0, \qquad |T_{1\bar2}| \le K+1,$$

and hence

$$\det T \ge (\Lambda a - K)\frac{\Lambda b}{2} - (K+1)^2 > 0.$$

Thus T is strictly positive on $U \setminus D$. A local weight $\varphi_0 + \Lambda s$ is psh off the analytic hypersurface D and locally bounded above near it. The removable-singularity theorem for psh functions extends it by the upper limit from the complement. Continuity identifies this extension explicitly:

$$\limsup_{\substack{(z',w')\to(z,0)\\ w'\neq 0}} (\varphi_0 + \Lambda s)(z', w') = \varphi_0(z, 0).$$

Hence $dd^c(\varphi_0 + \Lambda s) \ge 0$ on the entire chart. Finally, $s = |w|e^{\beta/2}$ is locally Lipschitz, proving the assertion. □

Proposition 2.5 (Global cusp metric)**.** *Let X be a smooth projective surface, $D \subset X$ a smooth connected curve with $D^2 < 0$, and H a line bundle such that $H|_D$ admits a smooth flat metric. Suppose*

$$A = H^{\otimes m} \otimes \mathcal{O}_X(-a_0 D)$$

is ample for some integers $m, a_0 > 0$. Then H admits a locally Lipschitz semipositive Hermitian metric, smooth with strictly positive curvature on $X \setminus D$.

Proof. Choose h_0, k, q, s globally as in Lemma 2.4, and write $\theta = \Theta_{h_0}(H)$ and $u_{\mathrm{loc}} = \Lambda s$. Let h_A be a smooth metric with $\omega_A = \Theta_{h_A}(A) > 0$. The identity of line bundles $H^{\otimes m} = A \otimes \mathcal{O}_X(a_0 D)$ defines a global smooth function G by

$$h_A \otimes k^{a_0} = h_0^m e^{-G}.$$

The divisor metric $h_{[D]} = ke^{-\log q}$ has curvature $[D]$. Thus the auxiliary metric and its weight relative to h_0 are

$$\begin{aligned} h_*^m = h_A \otimes h_{[D]}^{a_0}, \qquad h_* = h_0 e^{-\psi}, \qquad \psi = \frac{a_0}{m}\log q + \frac{G}{m}, \\ \theta + dd^c\psi = \frac{1}{m}\omega_A + \frac{a_0}{m}[D]. \end{aligned} \tag{2.12}$$

In particular, ψ is strictly θ-psh off D and tends uniformly to $-\infty$ on approach to D.

Fix $\varepsilon > 0$. Choose an even nonnegative smooth function χ_ε supported in $(-\varepsilon, \varepsilon)$ with integral one, and put

$$\rho_\varepsilon(t) = \int_{\mathbb{R}} |t - a|\chi_\varepsilon(a)\, da, \qquad \mathcal{M}_\varepsilon(x, y) = \frac{x + y + \rho_\varepsilon(x - y)}{2}.$$

Then $\rho_\varepsilon'' \geq 0$, $|\rho_\varepsilon'| \leq 1$, and $\rho_\varepsilon(t) = |t|$ for $|t| \geq \varepsilon$. Consequently $\mathcal{M}_\varepsilon$ is convex, nondecreasing in both variables, and

$$\mathcal{M}_\varepsilon(x+c, y+c) = c + \mathcal{M}_\varepsilon(x,y), \qquad \mathcal{M}_\varepsilon(x,y) = \max\{x,y\} \quad (|x-y| \geq \varepsilon).$$

For smooth functions p_1, p_2 and a tangent vector v,

$$\mathcal{L}_{\mathcal{M}_\varepsilon(p_1,p_2)}(v) = \sum_{j=1}^{2} (\mathcal{M}_\varepsilon)_j \mathcal{L}_{p_j}(v) + \sum_{j,k=1}^{2} (\mathcal{M}_\varepsilon)_{jk} \partial p_j(v) \overline{\partial p_k(v)}.$$

The second term is nonnegative, and the coefficients in the first term are nonnegative and sum to one. Thus strict plurisubharmonicity is preserved.

Choose $b > 0$ so that $\{q \leq 2b\} \subset U$, where the local metric is defined, and then choose $N \in \mathbb{R}$ and $0 < a < b/4$ such that

$$\begin{aligned} \psi + N &> u_{\mathrm{loc}} + 2\varepsilon \quad \text{on } \{b/2 \leq q \leq b\}, \\ \psi + N &< u_{\mathrm{loc}} - 2\varepsilon \quad \text{on } \{0 < q \leq 2a\}. \end{aligned} \tag{2.13}$$

The first inequality follows by taking N large on the compact shell; the second follows from

$$\frac{a_0}{m} \log(2a) + \max_X (G/m) + N < -2\varepsilon.$$

Define

$$h = h_0 e^{-u}, \qquad u = \begin{cases} u_{\mathrm{loc}}, & q \leq a, \\ \mathcal{M}_\varepsilon(u_{\mathrm{loc}}, \psi + N), & a < q < b, \\ \psi + N, & q \geq b. \end{cases} \tag{2.14}$$

The expressions coincide on collars of $q = a$ and $q = b$ by (2.13). On the annulus, a local weight is

$$\varphi_0 + u = \mathcal{M}_\varepsilon(\varphi_0 + u_{\mathrm{loc}}, \varphi_0 + \psi + N),$$

so it is smooth and strictly psh. The inner weight is psh by Lemma 2.4; the outer one is strictly psh by (2.12). The metric is locally Lipschitz near D and smooth with strictly positive curvature elsewhere. □

Proposition 2.6. *For the ruled surface (2.2), L admits a locally Lipschitz semipositive Hermitian metric, smooth with strictly positive curvature on $S \setminus Y$. Its curvature has bounded local potentials and minimal singularities.*

Proof. Equation (2.4) and Proposition 2.1 give

$$Y^2 = -d < 0, \qquad L|_Y \simeq \mathcal{O}_Y, \qquad L^{\otimes 2} \otimes \mathcal{O}_S(-Y) \text{ ample.}$$

Apply Proposition 2.5 with $(X, D, H, m, a_0) = (S, Y, L, 2, 1)$. Write the resulting curvature as $T = \theta + dd^c u \geq 0$, where u is continuous on compact S. For every global quasi-psh function $v \not\equiv -\infty$ with $\theta + dd^c v \geq 0$,

$$v \leq u + \sup_S v - \inf_S u.$$

This is the defining comparison for minimal singularities, and the local potentials $\varphi_0 + u$ are bounded. □

Remark 2.7. The choices $g = 2$, $F = \mathcal{O}_B(p)$ and $F = K_B$ yield respectively

$$(Y^2, L^2) = (-1, 1), \qquad (Y^2, L^2) = (-2, 2).$$

Proposition 2.6 gives bounded potentials with minimal singularities for both classes considered in [Zha25, Section 4]. Proposition 2.3 excludes C^1 semipositive weights in either case.

3. Exhaustions and the regularity criterion

We use the following regularity classes:

$$\begin{aligned}\mathcal{R} \in \{C^r : r \in \mathbb{Z}_{\geq 0} \cup \{\infty\}\} \cup \{C^\omega\} \\ \cup \{C^{k,\alpha}_{\mathrm{loc}} : k \in \mathbb{Z}_{\geq 0},\ 0 < \alpha \leq 1\}.\end{aligned} \tag{3.1}$$

Here $C^{k,\alpha}_{\mathrm{loc}}$ means C^k with locally α-Hölder derivatives of order k; $C^{k,1}_{\mathrm{loc}}$ is not identified with C^{k+1}. For every class in (3.1), the positive metric coefficients h_i, their reciprocals, and the weights $-\log h_i$ have the same regularity.

Lemma 3.1. *Let $A \to Z$ be a holomorphic line bundle over a compact complex manifold. If h is a semipositive Hermitian metric of class $\mathcal{R}$, then $\rho(v) = \|v\|^2_{h^{-1}}$ is a psh exhaustion of $\mathrm{Tot}(A^{-1})$ of the same class.*

Proof. Let e_i be a local frame of A over $B_i \subset Z$, with $\|e_i\|^2_h = e^{-\varphi_i}$. For $v = \lambda_i e_i^*$,

$$\rho(z, \lambda_i) = |\lambda_i|^2 e^{\varphi_i(z)}. \tag{3.2}$$

If $e_j = g_{ij} e_i$, then $\lambda_j = g_{ij}\lambda_i$ and $\varphi_j = \varphi_i - \log|g_{ij}|^2$, so

$$|\lambda_j|^2 e^{\varphi_j} = |g_{ij}\lambda_i|^2 e^{\varphi_i - \log|g_{ij}|^2} = |\lambda_i|^2 e^{\varphi_i}.$$

Thus (3.2) is global and belongs to $\mathcal{R}$, including at $\lambda_i = 0$.

To check positivity without second derivatives of φ_i, set

$$\rho_\varepsilon = (|\lambda_i|^2 + \varepsilon) e^{\varphi_i}, \qquad \varepsilon > 0.$$

The local function $\log \rho_\varepsilon = \log(|\lambda_i|^2 + \varepsilon) + \varphi_i$ is psh, since

$$\frac{\partial^2}{\partial\lambda_i \partial\bar{\lambda}_i} \log(|\lambda_i|^2 + \varepsilon) = \frac{\varepsilon}{(|\lambda_i|^2 + \varepsilon)^2} \geq 0.$$

Convex increasing composition makes ρ_ε psh. Moreover,

$$\sup_K |\rho_\varepsilon - \rho| \leq \varepsilon \sup_K e^{\varphi_i} \longrightarrow 0 \qquad (K \Subset A^{-1}|_{B_i}).$$

Passing to the limit in the submean inequalities proves that ρ is psh throughout the bundle chart.

Fix a smooth Hermitian metric k on A^{-1}. The ratio

$$a(z) = \frac{\|v\|^2_{h^{-1}}}{\|v\|^2_k}, \qquad 0 \neq v \in A^{-1}_z,$$

is independent of v and is a positive continuous function on Z. Compactness gives

$$0 < m = \min_Z a \leq \max_Z a = M < \infty, \qquad m\|v\|^2_k \leq \rho(v) \leq M\|v\|^2_k.$$

For $c \geq 0$, $\{\rho \leq c\}$ is a closed subset of the compact disk bundle $\{\|v\|^2_k \leq c/m\}$; for $c < 0$ it is empty. Hence ρ is an exhaustion. □

Lemma 3.2 (Compact-group averaging)**.** *If a compact group G acts continuously on a locally compact Hausdorff space and F_0 is a continuous exhaustion, then $F(x) = \int_G F_0(gx)\, dg$ is an exhaustion. For the fiberwise circle action on a holomorphic line bundle, averaging preserves plurisubharmonicity and every class in (3.1).*

Proof. Continuity follows from uniform continuity on compact orbit sets. For $K_c = \{F_0 \le c\}$,

$$\{F \le c\} \subset G \cdot K_c.$$

Indeed, $x \notin G \cdot K_c$ implies $\min_{g\in G} F_0(gx) > c$, and therefore $F(x) > c$. The set $G \cdot K_c$ is the continuous image of the compact set $G \times K_c$, so its closed subset $\{F \le c\}$ is compact.

For the circle action $R_\theta(v) = e^{i\theta}v$, put

$$F(v) = \frac{1}{2\pi}\int_0^{2\pi} F_0(R_\theta v)\, d\theta. \tag{3.3}$$

If f is a holomorphic disc defined near $\overline{\Delta}$, then

$$\begin{aligned} F(f(0)) &\le \frac{1}{(2\pi)^2}\int_0^{2\pi}\int_0^{2\pi} F_0(R_\theta f(e^{is}))\, ds\, d\theta \\ &= \frac{1}{2\pi}\int_0^{2\pi} F(f(e^{is}))\, ds. \end{aligned}$$

Thus F is psh, and changing the variable θ gives $F(R_\eta v) = F(v)$.

In real bundle coordinates $x = (z, \lambda)$, write $T_\theta x = (z, e^{i\theta}\lambda)$. The maps T_θ are linear with uniformly bounded norms. For $F_0 \in C^r$, $r \ge 1$, differentiation on compact coordinate sets yields

$$D^\beta F(x) = \frac{1}{2\pi}\int_0^{2\pi} D_x^\beta (F_0 \circ T_\theta)(x)\, d\theta, \qquad |\beta| \le r.$$

For $F_0 \in C^{k,\alpha}_{\mathrm{loc}}$, a finite cover of the compact orbit set gives, uniformly in θ,

$$|D^\beta(F_0 \circ T_\theta)(x) - D^\beta(F_0 \circ T_\theta)(y)| \le C|x-y|^\alpha, \qquad |\beta| = k.$$

Integration proves the same estimate for $D^\beta F$, including $k = 0$ and $\alpha = 1$.

For the analytic case, $\mathcal{G}(x,\theta) = F_0(T_\theta x)$ is jointly real analytic near $\{x_0\} \times S^1$. Complexify its convergent power series on finitely many angular charts covering S^1. After shrinking, their holomorphic extensions are bounded by one constant $M > 0$ on a common polydisc $|X - x_0|_\infty < 2R$, $R > 0$, in the complexified real variables, uniformly over the real angular parameters. Cauchy's estimates therefore give

$$\left|\frac{D_x^\beta \mathcal{G}(x_0,\theta)}{\beta!}\right| \le MR^{-|\beta|} \qquad (\beta \in \mathbb{Z}^m_{\ge 0},\ \theta \in S^1), \tag{3.4}$$

where m is the real coordinate dimension. For $|x - x_0|_\infty \le r < R$,

$$\sum_\beta \left|\frac{D_x^\beta \mathcal{G}(x_0,\theta)}{\beta!}(x-x_0)^\beta\right| \le M\sum_\beta (r/R)^{|\beta|} = M(1-r/R)^{-m}.$$

Termwise integration is justified uniformly in θ and gives

$$F(x) = \sum_\beta \left(\frac{1}{2\pi}\int_0^{2\pi} \frac{D_x^\beta \mathcal{G}(x_0,\theta)}{\beta!}\, d\theta\right)(x-x_0)^\beta.$$

This convergent power series proves $F \in C^\omega$. □

Proposition 3.3 (Recovery of the metric)**.** *Let $A \to Z$ be a holomorphic line bundle over a compact complex manifold. If $\mathrm{Tot}(A^{-1})$ admits a psh exhaustion of class $\mathcal{R}$ in (3.1), then A admits a semipositive Hermitian metric of the same class.*

Proof. Let F be the circle average (3.3) of the given exhaustion. For a local frame e_i of A over B_i, define

$$U_i(z,t) = F(e^{t/2}e_i^*(z)), \qquad V_i(z,\zeta) = F(e^{\zeta/2}e_i^*(z)).$$

Both parametrizations are real analytic, so U_i and V_i inherit the class $\mathcal{R}$. The second parametrization is holomorphic, so V_i is psh. Circle invariance gives

$$V_i(z,\zeta) = U_i(z,\operatorname{Re}\zeta), \qquad F(\lambda_i e_i^*(z)) = U_i(z,\log|\lambda_i|^2) \quad (\lambda_i \neq 0).$$

Fix z and write $u(t) = U_i(z,t)$. Subharmonicity in $\zeta = t + is$ implies, as distributions,

$$\Delta_\zeta u(\operatorname{Re}\zeta) = u''(t) \geq 0.$$

Testing against $\eta(t)\chi(s)$ with $\eta, \chi \geq 0$ and $\int \chi = 1$ gives $\langle u'', \eta\rangle \geq 0$. Convolution in t gives convex smooth functions converging locally uniformly to u; hence u is convex. Continuity at zero and properness of F give

$$\lim_{t\to-\infty} u(t) = F(0_z), \qquad \lim_{t\to+\infty} u(t) = +\infty. \tag{3.5}$$

If $u(b) < u(a)$ for $a < b$, the negative secant slope $d = (u(b) - u(a))/(b-a)$ would give

$$\frac{u(a)-u(x)}{a-x} \leq d < 0, \qquad u(x) \geq u(a) - d(a-x) \longrightarrow +\infty \quad (x \to -\infty),$$

contrary to (3.5). Thus u is nondecreasing.

Choose $c > \max_{z\in Z} F(0_z)$. The intermediate value theorem gives a point s with $u(s) = c$. If $u(s) = u(t) = c$ for $s < t$, then for $x < s$ the ordered secant slopes give

$$0 \leq \frac{u(s)-u(x)}{s-x} \leq \frac{u(t)-u(s)}{t-s} = 0.$$

This would give $u(x) = c$ for all $x < s$, again contradicting (3.5). Consequently there is a unique $\sigma_i(z)$ such that

$$U_i(z,\sigma_i(z)) = c, \qquad U_i(z,t) \leq c \iff t \leq \sigma_i(z). \tag{3.6}$$

In particular,

$$\{F < c\} \cap A^{-1}|_{B_i} = \{(z,\lambda_i) : |\lambda_i|^2 < e^{\sigma_i(z)}\}. \tag{3.7}$$

For $s_0 = \sigma_i(z_0)$ and $\varepsilon > 0$,

$$U_i(z_0, s_0 - \varepsilon) < c < U_i(z_0, s_0 + \varepsilon).$$

These strict inequalities persist near z_0 and force $s_0 - \varepsilon < \sigma_i(z) < s_0 + \varepsilon$. Thus σ_i is continuous.

We next prove that $-\sigma_i$ is psh without differentiating it. Let f be a holomorphic map from a neighborhood of $\overline{\Delta}$ to B_i. The harmonic extension of $\sigma_i \circ f|_{\partial\Delta}$ is

$$H(re^{it}) = \frac{1}{2\pi}\int_0^{2\pi} \frac{1-r^2}{1-2r\cos(t-\theta)+r^2}\sigma_i(f(e^{i\theta}))\,d\theta.$$

It is continuous on $\overline{\Delta}$. Since Δ is simply connected, there is a holomorphic g with $\operatorname{Re} g = H$. The function

$$q(\zeta) = F(e^{g(\zeta)/2}e_i^*(f(\zeta))) = U_i(f(\zeta), H(\zeta))$$

is subharmonic on Δ, and the second expression extends continuously to the boundary with value c. The maximum principle and (3.6) give

$$q \leq c \implies H \leq \sigma_i \circ f \implies -\sigma_i(f(0)) \leq \frac{1}{2\pi}\int_0^{2\pi} -\sigma_i(f(e^{i\theta}))\,d\theta.$$

Applying this to every smaller disc proves that $-\sigma_i$ is psh. Only H and $U_i(f,H)$ require continuous boundary values; no boundary extension of the harmonic conjugate is used.

If $e_j = g_{ij}e_i$, then $\lambda_j = g_{ij}\lambda_i$, and the radii of the same sublevel in (3.7) satisfy

$$e^{\sigma_j} = |g_{ij}|^2 e^{\sigma_i}, \qquad \sigma_j = \sigma_i + \log|g_{ij}|^2.$$

Thus

$$\psi_i = -\sigma_i, \qquad \psi_j = \psi_i - \log|g_{ij}|^2, \qquad \|e_i\|_h^2 = e^{-\psi_i} = e^{\sigma_i} \tag{3.8}$$

defines a continuous Hermitian metric on A with $\Theta_h(A) = dd^c\psi_i \geq 0$.

It remains to retain the regularity of F. For C^1 data, convexity gives, for any $a < \sigma_i(z)$,

$$U_{i,t}(z, \sigma_i(z)) \geq \frac{c - U_i(z,a)}{\sigma_i(z) - a} > 0. \tag{3.9}$$

The C^r implicit function theorem therefore gives $\sigma_i \in C^r$ for every finite $r \geq 1$ for which $F \in C^r$. Applying this for all r gives the smooth case. For analytic F, complexify $U_i(x,t) - c$ in real coordinates near $(x_0, \sigma_i(x_0))$. The map

$$(X,T) \longmapsto (X, \widetilde{U}_i(X,T) - c)$$

has Jacobian determinant $U_{i,t}(x_0, \sigma_i(x_0)) \neq 0$. Its holomorphic local inverse supplies $T = S(X)$; conjugation and uniqueness give $S(\bar{X}) = \overline{S(X)}$. Thus its restriction to the real coordinates is the real analytic function σ_i.

For $C^{0,\alpha}_{\mathrm{loc}}$ data, omit the index i and choose

$$a < b < \sigma(z_0) < d, \qquad U(z_0, a) < U(z_0, b).$$

The limits (3.5) permit this choice. By continuity, there are a relatively compact base ball B around z_0 and $m > 0$ such that

$$b < \sigma(z) < d, \qquad \frac{U(z,b) - U(z,a)}{b-a} \geq m \qquad (z \in \overline{B}).$$

The local Hölder regularity of U on a neighborhood of $\overline{B} \times [b,d]$ gives one constant C such that

$$|U(z,t) - U(w,t)| \leq C|z-w|^\alpha \qquad (z,w \in B,\ t \in [b,d]).$$

Ordered secant slopes then yield

$$U(z,t) - U(z,s) \geq m(t-s), \qquad b \leq s \leq t \leq d.$$

If $\sigma(z) \geq \sigma(w)$, this implies

$$\begin{aligned} m|\sigma(z) - \sigma(w)| &\leq U(z,\sigma(z)) - U(z,\sigma(w)) \\ &= U(w,\sigma(w)) - U(z,\sigma(w)) \leq C|z-w|^\alpha. \end{aligned}$$

Interchanging z, w gives the same bound in the other case. Hence

$$|\sigma(z) - \sigma(w)| \leq (C/m)|z-w|^\alpha, \qquad 0 < \alpha \leq 1. \tag{3.10}$$

This includes the Lipschitz endpoint.

Finally, suppose $U \in C^{k,\alpha}_{\mathrm{loc}}$ with $k \geq 1$. The classical theorem first gives $\sigma \in C^k$. On a smaller convex coordinate neighborhood, $x \mapsto (x, \sigma(x))$ is Lipschitz and $U_t(x,\sigma(x))$ is bounded below by a positive constant. Differentiating the level equation gives

$$\sigma_{x_j} = -\frac{U_{x_j}(x,\sigma(x))}{U_t(x,\sigma(x))}, \qquad U_t(x,\sigma(x))D^\beta\sigma = -P_\beta, \quad |\beta| = k,$$

where P_β is a finite sum of products of derivatives of U of order at most k, evaluated on the graph, and derivatives of σ of order at most $k-1$. For $k = 1$, the first formula has α-Hölder numerator and denominator. For $k \geq 2$,

$$D^j\sigma \in C^{0,1}_{\mathrm{loc}}\ (j < k), \qquad D^jU \in C^{0,1}_{\mathrm{loc}}\ (j < k), \qquad D^kU \in C^{0,\alpha}_{\mathrm{loc}}.$$

Composition with the Lipschitz graph and multiplication preserve these local Hölder estimates. Division by U_t therefore gives $D^\beta\sigma \in C^{0,\alpha}_{\mathrm{loc}}$, also when $\alpha = 1$. Formula (3.8) now proves the assertion in every class. □

Theorem 3.4 (Regularity criterion)**.** *Let $A \to Z$ be a holomorphic line bundle over a compact complex manifold. For each class $\mathcal{R}$ in (3.1),*

$$\mathrm{Tot}(A^{-1}) \text{ admits a psh exhaustion of class } \mathcal{R}$$
$$\Longleftrightarrow$$
$$A \text{ admits a Hermitian metric of class } \mathcal{R} \text{ with psh weights.}$$

No Kähler or projectivity assumption on Z is required.

Proof. The forward implication is Proposition 3.3; the reverse implication is Lemma 3.1. □

For C^2 data the two positivity arguments also have explicit Levi-form identities. Differentiating $U(z, \sigma(z)) = c$ twice gives

$$0 = U_{z_j\bar{z}_k} + U_{z_jt}\sigma_{\bar{z}_k} + U_{t\bar{z}_k}\sigma_{z_j} + U_{tt}\sigma_{z_j}\sigma_{\bar{z}_k} + U_t\sigma_{z_j\bar{z}_k}.$$

Since $V(z,\zeta) = U(z, \operatorname{Re}\zeta)$, $V_{z_j\bar{\zeta}} = U_{z_jt}/2$ and $V_{\zeta\bar{\zeta}} = U_{tt}/4$. Contracting the displayed identity with $\xi_j\bar{\xi}_k$ gives

$$\mathcal{L}_{-\sigma}(\xi) = \frac{\mathcal{L}_V(\xi, 2\partial\sigma(\xi))}{U_t} \geq 0. \tag{3.11}$$

For the dual norm, direct differentiation and completion of the square give

$$\mathcal{L}_{|\lambda|^2e^\varphi}(\xi,\eta) = e^\varphi\left(|\eta + \lambda\partial\varphi(\xi)|^2 + |\lambda|^2\mathcal{L}_\varphi(\xi)\right) \geq 0.$$

At $\lambda = 0$ this is $e^\varphi|\eta|^2$.

Proof of Theorem 1.1. Take the ruled surface S, the section Y, and the line bundle L constructed in Section 2, for example with $F = K_B$. The total space $M = \mathrm{Tot}(L^{-1})$ has the compactification

$$M = \mathbf{P}_{\mathrm{lines}}(\mathcal{O}_S \oplus L^{-1}) \setminus \mathbf{P}_{\mathrm{lines}}(L^{-1}), \qquad v \longmapsto [(1, v)]. \tag{3.12}$$

The inverse sends $[(a, b)]$, with $a \neq 0$, to b/a. Since S is projective and L is algebraic, the projective bundle in (3.12) is projective. Therefore $\dim_{\mathbb{C}} M = 3$, M is smooth and quasi-projective, and the restriction of a projective Kähler form makes it Kähler.

Let h be the locally Lipschitz semipositive metric of Proposition 2.6. Lemma 3.1 gives the locally Lipschitz psh exhaustion

$$\rho(v) = \|v\|^2_{h^{-1}}. \tag{3.13}$$

If M admitted a C^1 psh exhaustion, Theorem 3.4 would give a C^1 semipositive Hermitian metric on L, contrary to Proposition 2.3. □

4. A non-Kähler modification over one point

Proposition 4.1. *Let $f : X' \to X$ be a proper surjective holomorphic map of nonempty complex manifolds, biholomorphic outside $f^{-1}(K)$ for some compact $K \subset X$. For $r \in \mathbb{Z}_{\geq 0} \cup \{\infty\}$, X' admits a C^r psh exhaustion if and only if X does.*

Proof. If u is a C^r psh exhaustion of X, then $u \circ f$ is C^r and psh, and

$$\{u \circ f \leq a\} = f^{-1}(\{u \leq a\})$$

is compact by properness.

Conversely, let F be a C^r psh exhaustion of X'. Assume $K \neq \varnothing$, choose an open neighborhood $K \subset U \Subset X$, and take $c > \max_{f^{-1}(\overline{U})} F$. Define

$$\chi(t) = \begin{cases} 0, & t \leq c, \\ \displaystyle\int_c^t e^{-1/(s-c)}\, ds, & t > c. \end{cases}$$

The function χ is smooth, convex and nondecreasing, with $\chi(t) \to +\infty$ as $t \to +\infty$. Let $j : X \setminus K \to X' \setminus f^{-1}(K)$ be the holomorphic inverse of f. Then

$$G = \chi \circ F \circ j \quad \text{on } X \setminus K, \qquad G = 0 \quad \text{on } K$$

vanishes throughout U, because $F < c$ on $f^{-1}(U)$. It is therefore C^r across K. On $X \setminus K$, convex nondecreasing composition preserves plurisubharmonicity; for $r \geq 2$ this is the identity

$$\mathcal{L}_{\chi \circ F}(\xi) = \chi'(F)\mathcal{L}_F(\xi) + \chi''(F)|\partial F(\xi)|^2 \geq 0.$$

On U the function G is constant, so it is psh on all of X. For each $a \geq 0$, choose b such that $\chi(t) \leq a \Rightarrow t \leq b$. If $x \notin K$ and $G(x) \leq a$, then $F(j(x)) \leq b$. Hence

$$\{G \leq a\} \subset K \cup f(\{F \leq b\})$$

is a closed subset of a compact set and is compact. Negative sublevels are empty. If $K = \varnothing$, f is a biholomorphism and the inverse pullback gives the conclusion directly. □

Proposition 4.2. *For every point x of a connected smooth complex threefold X, there is a connected smooth non-Kähler threefold N and a proper modification $\mu : N \to X$ that is biholomorphic over $X \setminus \{x\}$.*

Proof. Blow up x:

$$b : W = \mathrm{Bl}_x X \to X, \qquad E = b^{-1}(x) \simeq \mathbf{P}^2.$$

In E, the curves

$$A = \{w = 0\}, \qquad B = \{uv = w^2\}$$

meet transversely at $p = [1 : 0 : 0]$ and $q = [0 : 1 : 0]$. Set $W_p = W \setminus \{q\}$ and $W_q = W \setminus \{p\}$. On W_p and W_q respectively, form

$$N_p = \mathrm{Bl}_{\widetilde{B \cap W_p}}(\mathrm{Bl}_{A \cap W_p} W_p),$$
$$N_q = \mathrm{Bl}_{\widetilde{A \cap W_q}}(\mathrm{Bl}_{B \cap W_q} W_q).$$

Here a tilde denotes strict transform. Over $W_p \cap W_q = W \setminus \{p, q\}$ the two centers are disjoint. The two blowup orders have a canonical isomorphism there, agreeing with the identity off $A \cup B$. Glue by this isomorphism to obtain N, with maps

$$\nu : N \longrightarrow W, \qquad \mu = b \circ \nu : N \longrightarrow X.$$

All centers, including their strict transforms, are smooth. Two points of N with distinct images have disjoint neighborhoods pulled back from W. Two points over the same image lie together in N_p or N_q, where they can be separated. Thus N is a smooth Hausdorff manifold. The map ν is proper because its restrictions over the open cover W_p, W_q are compositions of proper blowups. Hence μ is proper as well. The two connected blowup charts overlap, so N is connected. Every center lies in $E = b^{-1}(x)$; therefore μ is biholomorphic over $X \setminus \{x\}$.

We verify the non-Kähler obstruction, as in [Bar15, Section 6] and [Sta, Tag 08J1]. Let D_A, D_B be the exceptional surfaces dominating A, B, with the induced maps

$$f_A : D_A \longrightarrow A, \qquad f_B : D_B \longrightarrow B.$$

They are smooth compact surfaces: near either crossing they are ruled surfaces or the blowup of a ruled surface at one point, and they are closed in the proper inverse image of $A \cup B$. Let $\alpha, \beta \in H_2(N, \mathbb{R})$ be their general fiber classes. For any two points $a, a' \in A \simeq \mathbf{P}^1$, choose a rational function r with $\operatorname{div}(r) = [a] - [a']$. Then

$$\operatorname{div}(r \circ f_A) = f_A^*[a] - f_A^*[a'].$$

Thus all fiber divisors of f_A are linearly equivalent and represent α; the analogous assertion holds for f_B and β.

We compute the special fibers, including their multiplicities. Near p, take coordinates with $A = \{y = z = 0\}$ and $B = \{x = z = 0\}$. The first blowup has a chart

$$(x, y, t) \longmapsto (x, y, z = yt).$$

Its exceptional divisor is $y = 0$, and the strict transform of B is $x = t = 0$. A chart of the second blowup is

$$(r, y, t) \longmapsto (x = rt, y, t).$$

On the strict transform $D_A = \{y = 0\}$, the map f_A is $(r, t) \mapsto x = rt$. Its fiber over p is therefore

$$\operatorname{div}(rt) = \{r = 0\} + \{t = 0\} = R_p + T_p.$$

The closures of these two components are respectively the strict transform of the first exceptional fiber and the exceptional curve of its surface blowup. Both are smooth rational curves, and the factorization rt gives multiplicity one. The second exceptional surface D_B has fiber T_p over p. Interchanging A and B in the same calculation at q gives

$$\alpha = [R_p] + [T_p], \quad \beta = [T_p], \qquad \beta = [R_q] + [T_q], \quad \alpha = [T_q].$$

Subtracting the first pair and the second pair yields $[R_p] = \alpha - \beta$ and $[R_q] = \beta - \alpha$, hence

$$[R_p] + [R_q] = 0 \quad \text{in } H_2(N, \mathbb{R}). \tag{4.1}$$

The relations take place on compact surfaces, so they hold in ordinary homology even when N is noncompact. A Kähler form would give

$$0 = \int_{R_p} \omega + \int_{R_q} \omega > 0,$$

a contradiction. □

Proof of Theorem 1.2. Apply Proposition 4.2 to M at x, and let ρ be its locally Lipschitz psh exhaustion from Theorem 1.1. The pullback $\rho \circ \mu$ is a locally Lipschitz psh exhaustion because μ is locally Lipschitz and proper. A C^1 psh exhaustion on N would descend after truncation by Proposition 4.1, contradicting Theorem 1.1. □

5. Surface regularization and boundary criteria

Throughout this section, $X = S \setminus \operatorname{supp} D$, where S is a smooth compact connected complex surface and $D \neq 0$ is effective. We identify reduced divisors with their supports. A *divergent collar* is a psh function on a punctured neighborhood of $\operatorname{supp} D$ tending uniformly to $+\infty$ at the boundary. An exhaustion of X always has this divergence property.

5.1. **Boundary tools.**

Lemma 5.1. (i) *If a connected component of* $\operatorname{supp} D$ *has negative definite intersection matrix,* X *has no continuous psh exhaustion.*

(ii) *Smooth divergent psh collars at all connected boundary components give a smooth psh exhaustion of* X*.*

(iii) *For a connected reduced divisor* $E = \bigcup E_i$ *with simple normal crossings (SNC), let* λ *be the largest eigenvalue of* $I = (E_i \cdot E_j)$*. If* $\lambda > 0$*,* E *has a smooth strictly psh divergent collar. If* $\lambda = 0$*,* $I \le 0$*,* $\dim \ker I = 1$*, and*

$$A = \sum_i a_i E_i, \qquad a_i \in \mathbb{Z}_{>0}, \qquad A \cdot E_i = A^2 = 0. \tag{5.1}$$

When S *is projective,* A *is nef.*

Proof. For (i), let E be the boundary component. Grauert's theorem [Gra62, §4, subsection 8(e)] contracts E to a point p of a normal surface Z. Write the contraction as $q : W \to Z$; its restriction $W \setminus E \to Z \setminus \{p\}$ is a biholomorphism. An exhaustion ρ on X would therefore give $u = \rho \circ (q|_{W\setminus E})^{-1}$, psh and divergent at p. Choose a finite local projection

$$\pi : V \longrightarrow \Delta^2, \qquad \pi^{-1}(0) = \{p\}, \qquad v(z) = \max_{y \in \pi^{-1}(z)} u(y).$$

The function v is continuous on $\Delta^2 \setminus \{0\}$. Off the branch locus it is a finite maximum of psh functions. Properness of π gives local boundedness at each remaining point away from 0, so removability makes v psh there as well. Fix $0 < r < 1$ and set

$$M_r = \max_{\{|z_1| \le r,\ |z_2| = r\}} v < \infty.$$

For $0 < |z_1| < r$, the subharmonic maximum principle on the vertical disc gives $v(z_1, z_2) \le M_r$ for $|z_2| < r$. Continuity gives the same bound when $z_1 = 0$ and $z_2 \neq 0$. Hence $u \le M_r$ near p, contradicting its divergence.

For (ii), choose disjoint neighborhoods W_E of the boundary components so that each collar Φ_E is defined on a neighborhood of $\overline{W}_E \setminus E$. Choose $T_E > \sup_{\partial W_E} \Phi_E$ and a smooth convex nondecreasing χ_E, zero on $(-\infty, T_E]$ and tending to $+\infty$. Define

$$\rho_E(x) = \begin{cases} \chi_E(\Phi_E(x)), & x \in W_E \setminus E, \\ 0, & x \notin W_E. \end{cases}$$

The two formulas agree on a neighborhood of ∂W_E, and

$$\mathcal{L}_{\chi_E \circ \Phi_E}(\xi) = \chi_E'(\Phi_E)\mathcal{L}_{\Phi_E}(\xi) + \chi_E''(\Phi_E)|\partial \Phi_E(\xi)|^2 \ge 0.$$

Thus $\rho = \sum_E \rho_E$ is smooth and psh. Every sequence leaving compact subsets of X approaches D along a subsequence, where the corresponding summand tends to infinity. This proves properness.

For (iii), Perron–Frobenius applied to $I + c\mathrm{Id}$ shows that λ is simple and has a strictly positive eigenvector v. When $\lambda > 0$, both $v > 0$ and $Iv > 0$ persist under small perturbations. Approximate v by a rational vector and clear denominators to obtain $A = \sum a_i E_i$ with $a_i \in \mathbb{Z}_{>0}$ and $A \cdot E_i > 0$ for all i. Thus $\mathcal{O}_S(A)|_E$ is ample. A projective embedding by a power gives a Fubini–Study metric on this restriction. Extend its weights locally to S and patch them relative to a reference metric. The extensions have positive tangential curvature along E; near a node, lift the embedding sections from E to a coordinate neighborhood in S. Their ratios have differentials spanning the two-dimensional Zariski cotangent space of the node, and hence the ambient cotangent space. Their Fubini–Study weights have positive curvature in both directions.

Keep these weights on smaller node neighborhoods when patching. The other extensions agree along E, so the patched metric retains positive tangential curvature at its smooth points.

Let $q = \|s_E\|^2$ for a smooth metric on $\mathcal{O}_S(E)$. At a smooth point of $E = \{w = 0\}$, write $q = |w|^2 e^{-\gamma}$. Then

$$\partial\bar\partial q|_E = e^{-\gamma}|_E \, dw \wedge d\bar w.$$

Away from the node neighborhoods the curvature matrix of the patched metric has the form $\left(\begin{smallmatrix} a & b \\ \bar b & d \end{smallmatrix}\right)$ in tangential and normal directions, with $a \geq \varepsilon > 0$. Adding Cq to its weights replaces d by $d + Ce^{-\gamma}$ along E; compactness permits one C for which $a(d + Ce^{-\gamma}) > |b|^2$. At the nodes $dd^c q = 0$, and at other points of E it is semipositive, so positivity is preserved there. Continuity now gives a metric h of positive curvature near all of E. On the complement,

$$dd^c(-\log \|s_A\|_h^2) = \Theta_h(\mathcal{O}_S(A)) > 0,$$

and $a_i > 0$ gives uniform divergence. For $\lambda = 0$, every other eigenvalue is negative. Since I has integer entries and rank one less than its size, its positive kernel has a rational generator; clearing denominators proves (5.1). Effectiveness gives $A \cdot C \geq 0$ for every curve $C \not\subset E$, which proves nefness in the projective case. □

For a boundary resolution $p : \widetilde S \to S$, the restriction $\widetilde S \backslash p^{-1}(\operatorname{supp} D) \to X$ is a biholomorphism. Interior blowups preserve both exhaustion properties by Proposition 4.1.

Proposition 5.2. *A connected noncompact complex surface X with a continuous psh exhaustion and $\mathcal{O}(X) \neq \mathbb{C}$ has a smooth psh exhaustion.*

Proof. Vîjîitu's theorem [Vîj22a, Theorem 1.1] gives holomorphic convexity. Let $r : X \to Y$ be the proper Remmert reduction and let τ be a smooth strictly psh exhaustion of the Stein space Y. Then

$$\rho = \tau \circ r, \qquad \{\rho \leq c\} = r^{-1}(\{\tau \leq c\}) \Subset X$$

is a smooth psh exhaustion. □

Proposition 5.3. *For a line bundle A on a compact Riemann surface B,*

$$\operatorname{Tot}(A^{-1}) \text{ continuously weakly complete}$$
$$\iff \quad \deg A \geq 0 \quad \iff$$
$$\operatorname{Tot}(A^{-1}) \text{ smoothly weakly complete.}$$

Proof. For $d = \deg A \geq 0$, choose an area form ω of integral one and solve $dd^c f = d\omega - \theta_0$, where θ_0 is the curvature of a reference metric h_0. The right-hand side has integral zero, so the equation is solvable on the compact curve. The metric $h = e^{-f} h_0$ satisfies

$$\Theta_h(A) = \theta_0 + dd^c f = d\omega \geq 0,$$

and Lemma 3.1 applies. The map $v \mapsto \mathbb{C}(1, v)$ embeds $\operatorname{Tot}(A^{-1})$ into $\mathbf{P}_{\text{lines}}(\mathcal{O}_B \oplus A^{-1})$, with complement $Y_\infty = \mathbf{P}(A^{-1})$. Its normal bundle is

$$N_{Y_\infty/\mathbf{P}(\mathcal{O}_B \oplus A^{-1})} = \operatorname{Hom}(A^{-1}, \mathcal{O}_B) = A.$$

If $d < 0$, this is a negative-square boundary, contradicting Lemma 5.1(i). □

5.2. **Current averaging and regularization.**

Theorem 5.4. *Let S be compact Kähler and $D \neq 0$ a reduced divisor. If $X = S \setminus D$ has a continuous psh exhaustion, it has a smooth one. If X is not a modification of a Stein space, there is*

$$A = \sum_i a_i D_i, \qquad a_i \in \mathbb{Z}_{>0}, \qquad A \cdot D_i = A^2 = 0,$$

such that $\mathcal{O}_S(A)$ has a smooth Hermitian metric with zero curvature near D. Thus the entire boundary has a smooth pluriharmonic divergent collar.

Lemma 5.5. *Let (S, ω) be a compact Kähler surface and $X \subset S$ a connected noncompact open set with a continuous psh exhaustion ρ. If X is not a modification of a Stein space, there is a positive $(1,1)$-current T on S such that*

$$dd^c T = 0, \qquad \int_S T \wedge \omega = 1, \qquad \operatorname{supp} T \Subset X, \qquad \nu(T, x) = 0 \quad (x \in S). \tag{5.2}$$

Proof. Let Σ_X^0 be the intersection of the loci where continuous psh exhaustions fail to be strongly psh. By [MS20, Lemma 2.1],

$$(t_0, \infty) \subset \rho(\Sigma_X^0) \tag{5.3}$$

for some t_0. For $t > t_0$, no smooth strictly psh function exists near $K_t = \{\rho = t\}$. Indeed, if v were such a function, properness permits $\delta > 0$ with the compact shell $\{t - \delta \leq \rho \leq t + \delta\}$ contained in its domain. After scaling and adding a constant, arrange $0 < v < 1$ there. For $0 < \eta < 1$, $0 < \varepsilon < \eta\delta/2$, set $g(s) = s + \eta|s - t|$. The function

$$\psi = \begin{cases} \max\{g(\rho), \rho + \varepsilon v\}, & |\rho - t| < \delta, \\ g(\rho), & |\rho - t| \geq \delta \end{cases}$$

is well defined: near $|\rho - t| = \delta$, $g(\rho) - \rho = \eta|\rho - t| > \varepsilon v$, so the formulas agree. Both entries of the maximum are psh, and g is convex and increasing. On K_t the second entry strictly dominates, so $\psi = \rho + \varepsilon v$ is strongly psh near K_t. Finally $\psi \geq g(\rho)$ makes ψ an exhaustion. This contradicts $K_t \cap \Sigma_X^0 \neq \varnothing$.

Let $\mathcal{B}_t$ be the Banach space of continuous real $(1,1)$-forms on K_t, with the uniform norm induced by ω. Consider the subspace and open cone

$$V_t = \{dd^c f|_{K_t} : f \in C^\infty(S, \mathbb{R})\}, \qquad P_t = \{\eta \in \mathcal{B}_t : \eta > 0 \text{ on } K_t\}.$$

If $V_t \cap P_t$ were nonempty, continuity would make the corresponding f strictly psh on a neighborhood of K_t. Conversely, such a local function extends to S after multiplication by a cutoff equal to one near K_t. Hahn–Banach separation thus gives a nonzero continuous functional ℓ_t with

$$\ell_t|_{V_t} = 0, \qquad \ell_t(\eta) \geq 0 \quad (\eta \geq 0).$$

The second assertion follows by applying the separating inequality to $\eta + \varepsilon\omega$ and letting $\varepsilon \downarrow 0$. Moreover $\ell_t(\omega) > 0$: otherwise $-C\omega \leq \eta \leq C\omega$ would force $\ell_t(\eta) = 0$ for every η. Define the current by

$$\langle T_t, \eta \rangle = \frac{\ell_t(\eta|_{K_t})}{\ell_t(\omega|_{K_t})}.$$

Its positivity, support, and normalization are immediate; and $\langle dd^c T_t, f \rangle = \langle T_t, dd^c f \rangle = 0$ gives

$$T_t \geq 0, \qquad dd^c T_t = 0, \qquad \operatorname{supp} T_t \subset K_t, \qquad \int_S T_t \wedge \omega = 1. \tag{5.4}$$

Fix $t_0 < a < b$ and choose

$$t_{n,j} = a + \frac{j - 1/2}{n}(b-a), \qquad T_n = \frac{1}{n}\sum_{j=1}^{n} T_{t_{n,j}}.$$

These currents have mass one and support in the fixed compact shell $K = \rho^{-1}([a,b]) \Subset X$. A subsequence converges weakly to a positive current T. Testing against ω, against $dd^c f$, and against forms supported outside K respectively gives

$$\int_S T \wedge \omega = 1, \qquad dd^c T = 0, \qquad \operatorname{supp} T \subset K.$$

Fix a coordinate ball $B(x, 2R) \Subset X$, and put $\beta = dd^c|z|^2$. Lelong–Jensen monotonicity for positive dd^c-closed currents [DNS22, Appendix B, (B.1)–(B.2)] gives, uniformly in t, for $0 < r < R$,

$$r^{-2}\int_{B(x,r)} T_t \wedge \beta \le R^{-2}\int_{B(x,R)} T_t \wedge \beta \le C_R. \tag{5.5}$$

Indeed, $\beta \le c_R\omega$ on $B(x,R)$, so one may take $C_R = c_R/R^2$, independently of t. Put $m_r = \inf_{B(x,r)} \rho$ and $M_r = \sup_{B(x,r)} \rho$. Since $\operatorname{supp} T_t \subset \{\rho = t\}$, only indices with $t_{n,j} \in [m_r, M_r]$ contribute. Their number $N_n(r)$ satisfies

$$N_n(r) \le \frac{n(M_r - m_r)}{b-a} + 2.$$

Summing (5.5) over these indices gives

$$r^{-2}\int_{B(x,r)} T_n \wedge \beta \le C_R\left(\frac{\operatorname{osc}_{B(x,r)} \rho}{b-a} + \frac{2}{n}\right). \tag{5.6}$$

Multiply β by a cutoff equal to one on $B(x,R)$. The corresponding positive measures μ_n converge weakly to μ. For the open ball, the Portmanteau inequality and (5.6) give

$$\mu(B(x,r)) \le \liminf_n \mu_n(B(x,r)) \le \frac{C_R r^2}{b-a}(M_r - m_r).$$

Consequently,

$$r^{-2}\int_{B(x,r)} T \wedge \beta \le \frac{C_R}{b-a}\operatorname{osc}_{B(x,r)} \rho \longrightarrow 0 \qquad (r \downarrow 0). \tag{5.7}$$

Thus $\nu(T,x) = 0$ on X; outside X, T vanishes locally. □

Lemma 5.6. *For a positive dd^c-closed $(1,1)$-current T on a compact Kähler surface with all Lelong numbers zero, there is a class $\alpha_T \in H^{1,1}(S,\mathbb{R})$ satisfying*

$$\alpha_T \cdot [\eta] = \int_S T \wedge \eta \qquad (d\eta = 0), \tag{5.8}$$

for every smooth real $(1,1)$-form η. Moreover,

$$\alpha_T^2 \ge 0, \qquad dT \ne 0 \implies \alpha_T^2 > 0. \tag{5.9}$$

If $dT = 0$, then $\alpha_T = [T]$.

Proof. The decomposition [DNS22, §2.1 and Lemma B.2]

$$T = \Omega + \partial R + \bar\partial\overline{R} + i\partial\bar\partial u$$

has Ω smooth, real, and closed, R of bidegree $(0,1)$, and u a real distribution. For $d\eta = 0$, its pure bidegree implies $\partial\eta = \bar\partial\eta = 0$. Thus

$$\langle \partial R, \eta\rangle = \langle \bar\partial\overline{R}, \eta\rangle = \langle i\partial\bar\partial u, \eta\rangle = 0,$$

which proves (5.8) with $\alpha_T = [\Omega]$. The nondegenerate intersection pairing on $H^{1,1}(S,\mathbb{R})$ makes the class independent of the decomposition. The density formula [DNS22, Theorem 2.2(2), Corollary 2.4] gives

$$\alpha_T^2 = \|\vartheta\| + 2E(T), \qquad \|\vartheta\| \geq 0, \quad E(T) \geq 0, \quad E(T) = 0 \Longleftrightarrow dT = 0.$$

Here ϑ is the positive measure on the diagonal determined by a tangent current of $T \otimes T$, and $E(T) = \int_S \bar\partial R \wedge \partial\overline{R}$ is the current energy, independent of the chosen decomposition. The mass hypotheses hold because all Lelong numbers vanish. This proves (5.9). □

Proof of Theorem 5.4. If $r : X \to Y$ is a proper modification of a Stein space, the pullback of a smooth strictly psh exhaustion of Y is the required exhaustion. Otherwise the preceding lemmas give a current T with support in $K \Subset X$ and a class $\alpha = \alpha_T$ satisfying

$$\alpha \cdot [\omega] = 1, \qquad \alpha^2 \geq 0, \qquad dT \neq 0 \Longrightarrow \alpha^2 > 0. \tag{5.10}$$

For each component D_i, choose a metric of curvature θ_i and canonical section s_i. Choose $\chi \in C_c^\infty(X)$ equal to one near K, and extend $v_i = \chi \log \|s_i\|^2$ by zero to S. Since s_i is nonvanishing near K, Poincaré–Lelong gives $dd^c v_i = -\theta_i$ there. Therefore

$$\alpha \cdot [D_i] = \int_S T \wedge \theta_i = -\langle T, dd^c v_i\rangle = 0. \tag{5.11}$$

For a connected boundary component E, let λ_E be the largest eigenvalue of its intersection matrix I_E. Lemma 5.1(i) excludes $\lambda_E < 0$. If $\lambda_E > 0$, rational approximation gives an integral divisor $B = \sum b_i D_i$, with all $b_i > 0$ and $B \cdot D_i > 0$ on E. In particular, $B^2 = \sum b_i (B \cdot D_i) > 0$. Then $\alpha \cdot [B] = 0$, and Hodge index forces $\alpha^2 < 0$, contrary to (5.10). Therefore $\lambda_E = 0$. Perron–Frobenius gives a one-dimensional positive rational kernel of I_E. Choose a positive integral generator for each E and sum:

$$A = \sum_i a_i D_i, \qquad a_i > 0, \qquad \operatorname{supp} A = D, \qquad A \cdot D_i = A^2 = 0. \tag{5.12}$$

Here distinct irreducible curves have nonnegative intersection, and connectedness makes each matrix irreducible; these are the only properties of the components used in the matrix argument.

Set $c = (\int_A \omega)^{-1}$ and $v = \alpha - c[A]$. Effectiveness gives $\int_A \omega > 0$, while (5.10)–(5.11) give

$$v \cdot [\omega] = 0, \qquad v \cdot [A] = 0, \qquad v^2 = \alpha^2 \geq 0.$$

Hodge index is negative definite on $[\omega]^\perp$, so $v = 0$ and $\alpha^2 = 0$. The strict inequality in (5.10) then forces $dT = 0$. We have proved

$$dT = 0, \qquad [T] = \alpha = c[A], \qquad c = \left(\int_A \omega\right)^{-1} > 0. \tag{5.13}$$

The dd^c-lemma for currents now supplies a real distribution u with

$$dd^c u = T - c[A]. \tag{5.14}$$

Choose a neighborhood U of D disjoint from K. On a trivializing chart $U_i \subset U$, write $s_A = f_i e_i$. Poincaré–Lelong and (5.14) give the distributional identity

$$dd^c(u + c\log|f_i|^2) = -c[A] + c[A] = 0.$$

Elliptic regularity for pluriharmonic distributions yields

$$u = -c\log|f_i|^2 + h_i, \qquad h_i \text{ smooth and pluriharmonic on } U_i. \tag{5.15}$$

Indeed, taking a local trace gives a harmonic distribution, hence a smooth function, and the original $\partial\bar\partial$ equation then makes it pluriharmonic. The functions $-c\log|f_i|^2 + h_i$ agree on overlaps off D, since they represent the same distribution u. They therefore define a smooth pluriharmonic function on $U \setminus D$. In a finite relatively compact chart cover of D, each h_i is bounded and f_i tends to zero along D. Since every $a_i > 0$, this proves uniform divergence, including at singular points of the boundary. Lemma 5.1(ii) gives the exhaustion.

For the metric statement, set $b_i = u/c + \log|f_i|^2 = h_i/c$ on U_i. If $e_j = g_{ij}e_i$, then $f_j = g_{ij}^{-1}f_i$ and

$$b_j = b_i - \log|g_{ij}|^2, \qquad dd^c b_i = 0.$$

Thus $\|e_i\|^2_{h_{\mathrm{flat}}} = e^{-b_i}$ defines a smooth flat Hermitian metric on $\mathcal{O}_S(A)|_U$, with $-\log\|s_A\|^2_{h_{\mathrm{flat}}} = u/c$. For a global smooth metric h_0 with weights b_i^0, the differences $b_i - b_i^0$ define a single smooth function b on U. Choose a cutoff χ supported in U and equal to one on a smaller neighborhood of D. The global metric $h = e^{-\chi b}h_0$ agrees with h_{flat} there, proving the claimed curvature statement. □

Corollary 5.7. *Every smooth complex quasi-projective surface with a continuous psh exhaustion has a smooth psh exhaustion.*

Proof. Resolve a projective closure and principalize the complement ideal, with centers outside X [Wło05, Theorems 1.0.1–1.0.2]. The resulting morphism $p : S \to \overline{X}$ is an isomorphism over X, and $D = (p^{-1}(\overline{X}\setminus X))_{\mathrm{red}}$ is SNC. Thus p identifies $S\setminus D$ with X, including when the original boundary had isolated points. Apply Theorem 5.4 to each noncompact component. Compact components need only constant functions, and there are finitely many components. □

The case $\mathcal{O}(X) = \mathbb{C}$ occurs: take $X = \mathrm{Tot}(A^{-1})$ for non-torsion $A \in \mathrm{Pic}^0(B)$. Its squared unitary-flat norm is a smooth psh exhaustion. The degree-n fiberwise Taylor coefficient of $f \in \mathcal{O}(X)$ lies in $H^0(B, A^n) = 0$ for $n > 0$; the constant coefficient is constant on B. Thus $\mathcal{O}(X) = \mathbb{C}$.

Corollary 5.8. *Let $C \subset S$ be a smooth connected square-zero curve in a compact Kähler surface. A continuous divergent psh collar at C implies a smooth psh exhaustion of $S \setminus C$. If the complement is not a modification of a Stein space, $\mathcal{O}_S(C)$ is unitary flat near C and admits a pluriharmonic divergent collar.*

Proof. Let Φ be the collar, choose a neighborhood W of C such that Φ is defined near $\overline{W}\setminus C$, and take $M > \sup_{\partial W}\Phi$. Set

$$\rho(x) = \begin{cases} \max\{\Phi(x), M\}, & x \in W \setminus C, \\ M, & x \notin W. \end{cases}$$

The formulas agree near ∂W, and divergence at C makes ρ a continuous psh exhaustion. In Theorem 5.4, the single-component kernel generator can be chosen to be 1, so $A = C$. □

For reducible D, the flat metric is on the weighted bundle $\mathcal{O}_S(A)$; the kernel multiplicities need not all equal one.

5.3. **Numerical existence criteria.**

Theorem 5.9. *For a nonprojective S and an effective $D \neq 0$, the following are equivalent:*

(i) $S \setminus \operatorname{supp} D$ *is continuously weakly complete;*
(ii) $S \setminus \operatorname{supp} D$ *is smoothly weakly complete;*
(iii) *no connected boundary component has negative definite intersection matrix.*

The Hopf case uses [CFS+26, Theorem 1.6].

Proof. Only (iii)⇒(ii) requires proof. The classification facts used below are recalled in [FN05, §2]. If $a(S) = 1$, the algebraic reduction is a holomorphic elliptic fibration $f : S \to B$ with connected fibers, and every curve is vertical. For a fiber $F = \sum m_i C_i$, the identities $F \cdot C_i = 0$ imply $m_i C_i^2 = -\sum_{j \neq i} m_j (C_i \cdot C_j)$. Substitution gives

$$x^{\mathsf{T}} I x = -\sum_{i<j} (C_i \cdot C_j) m_i m_j \left(\frac{x_i}{m_i} - \frac{x_j}{m_j} \right)^2, \qquad I = (C_i \cdot C_j). \tag{5.16}$$

The right-hand side vanishes precisely when all x_i/m_i are equal, since the fiber is connected. Extend a vector on a proper subset of the components by zero; equality then forces the entire vector to vanish. Thus every proper principal submatrix is negative definite. Condition (iii) implies that the boundary contains each fiber support it meets, and $X = f^{-1}(B \setminus P)$ for a finite nonempty P. For a smooth subharmonic exhaustion τ of $B \setminus P$, $\tau \circ f$ is psh and $\{\tau \circ f \leq c\} = f^{-1}(\{\tau \leq c\})$ is compact.

Suppose $a(S) = 0$. By [FN05, Lemma 2.1], $c_1(L)^2 \leq 0$ for every line bundle L. Perron–Frobenius and (iii) therefore give an effective A of full boundary support with $A^2 = 0$. Indeed, a positive largest eigenvalue would give an effective divisor of positive square, and a negative one is excluded by (iii). At a blowdown $q : T \to T'$ with exceptional curve E, put $A' = q_* A$ and write $A = q^* A' + bE$. Since $q^* A' \cdot E = 0$ and $E^2 = -1$,

$$0 = A^2 = (A')^2 - b^2, \qquad (A')^2 \leq 0,$$

so $b = 0$ and $A = q^* A'$. In particular A' cannot vanish unless $A = 0$. Iterating to the minimal model $p : S \to S_0$ gives $A = p^* A_0 \neq 0$, $A_0^2 = 0$, and $\operatorname{supp} A = p^{-1}(\operatorname{supp} A_0)$. The restriction

$$p : S \setminus \operatorname{supp} A \longrightarrow S_0 \setminus \operatorname{supp} A_0$$

is proper because the deleted set is the full inverse image of $\operatorname{supp} A_0$. Pullback therefore preserves an exhaustion.

If S_0 is Kähler, it is a torus or K3 of algebraic dimension zero. On a torus, translating an irreducible curve shows its square is nonnegative, hence zero. Adjunction then gives arithmetic genus one. A rational normalization is impossible, since any map from $\mathbf{P}^1$ lifts to $\mathbb{C}^2$ and is constant. The curve is therefore a smooth elliptic curve, and a translate is an elliptic subtorus. The quotient onto the one-dimensional torus gives a nonconstant meromorphic function. On a K3, $h^0(\mathcal{O}(-A_0)) = 0$ because A_0 is effective and nonzero, while Riemann–Roch and Serre duality give

$$h^0(\mathcal{O}(A_0)) - h^1(\mathcal{O}(A_0)) = 2 + \tfrac{1}{2} A_0^2 = 2.$$

Two independent sections have a nonconstant meromorphic ratio. Both possibilities contradict $a(S_0) = 0$.

Otherwise S_0 is of class VII. If $b_2(S_0) = n > 0$, Enoki's square-zero-divisor theorem [Eno81, Main Theorem, p. 454] makes S_0 an Enoki surface and gives

$$A_0 = m C_{\mathrm{E}}, \qquad m \in \mathbb{Z}_{>0}, \qquad \operatorname{supp} A_0 = C_{\mathrm{E}},$$

where C_{E} is its distinguished reduced cycle; effectiveness of A_0 forces $m > 0$. By [Eno81, Proposition 3.5(iii), p. 460], $\mathcal{A} = S_0 \setminus C_{\mathrm{E}}$ is an affine-line bundle over an elliptic curve B_{E}. Its linearization Q has $\deg Q = -n$; this is the convention for the degree of an affine bundle in [Eno81, §2, p. 457].

Write its affine coordinate changes as $w_i = a_{ij}w_j + b_{ij}$. Extending these maps to $\mathbf{P}^1$ gives a compact ruled surface $\overline{\mathcal{A}}$ with infinity section Y_∞. In the coordinates $u_i = w_i^{-1}$ near this section,

$$u_i = \frac{u_j}{a_{ij} + b_{ij}u_j} = a_{ij}^{-1}u_j + O(u_j^2), \qquad N_{Y_\infty/\overline{\mathcal{A}}} \simeq Q^{-1}.$$

Consequently

$$Y_\infty^2 = \deg Q^{-1} = n > 0, \qquad \overline{\mathcal{A}} \setminus Y_\infty \simeq S_0 \setminus \operatorname{supp} A_0.$$

Lemma 5.1(iii) supplies a smooth strictly psh divergent collar at Y_∞, and (ii) gives a smooth psh exhaustion of its complement. If $b_2(S_0) = 0$, a minimal class VII surface containing a curve is elliptic or Hopf [Eno81, p. 453]; $a(S_0) = 0$ leaves Hopf. Since $b_2 = 0$, every curve has square zero and distinct curves have intersection zero. Adjunction gives arithmetic genus one, while rational normalizations are impossible because maps $\mathbf{P}^1 \to S_0$ lift to $\mathbb{C}^2 \setminus \{0\}$ and are constant. Thus the curves are disjoint smooth elliptic curves. Finite Ueda type gives a smooth strictly psh collar [CLPT18, Theorem 2.5]. Infinite type gives a unitary flat neighborhood by [CFS+26, Theorem 1.6], hence a pluriharmonic collar. Glue the collars. □

Theorem 5.10. *Let $f : S \to B$ be a smooth $\mathbf{P}^1$-bundle over a compact Riemann surface, let F be a fiber, and let $D \neq 0$ be a reduced SNC divisor. The three conditions in Theorem 5.9 are equivalent, with no restriction on $D \cdot F$.*

Proof. Necessity and positive-eigenvalue boundary components are handled by Lemma 5.1. For a connected component E with zero largest eigenvalue, choose its positive kernel divisor A. If E is vertical, it is a whole fiber, and the punctured-base construction applies. Otherwise $a = A \cdot F > 0$. A vertical component would be a fiber and have intersection a with A, contradicting $A \cdot E_i = 0$. Thus every E_i is horizontal. The ruling is projective; choose a section C_0 with $C_0^2 = -e$. Write

$$A \equiv aC_0 + bF, \qquad E_i \equiv d_iC_0 + b_iF, \qquad d_i = E_i \cdot F > 0.$$

Using $C_0 \cdot F = 1$ and $F^2 = 0$, compute

$$0 = A^2 = -ea^2 + 2ab, \qquad 0 = A \cdot E_i = ab_i + d_ib - ead_i.$$

Since $a > 0$, these identities give

$$b = ae/2, \qquad E_i \equiv d_i(C_0 + (e/2)F), \qquad E_i^2 = E_i \cdot E_j = 0 \quad (i \neq j).$$

Distinct components are therefore disjoint, and connectedness makes $E = C$ a single curve, smooth by the SNC hypothesis. For $d = C \cdot F$, the canonical class formula

$$K_S \equiv -2C_0 + (2g(B) - 2 - e)F$$

and adjunction give $2g(C)-2 = (K_S+C)\cdot C = d(2g(B)-2)$. The nonconstant map $f|_C : C \to B$ is finite of degree d. Riemann–Hurwitz now gives degree zero for its effective ramification divisor, so it is étale.

For $d = 1$, the complement is an affine-line bundle with degree-zero linearization. In unitary flat frames, initially write the affine transitions as $w_i = a_{ij}w_j + b_{ij}(z)$ with $|a_{ij}| = 1$. The construction of [KU25, Proposition 2.1] gives pluriharmonic functions h_i with $h_i = a_{ij}h_j + b_{ij}$.

On simply connected charts split $h_i = f_i + g_i$, with f_i holomorphic and g_i antiholomorphic. The coordinate changes $w_i \mapsto w_i - f_i$ replace b_{ij} by

$$c_{ij} = b_{ij} + a_{ij} f_j - f_i = g_i - a_{ij} g_j.$$

This expression is both holomorphic and antiholomorphic, hence constant. In the new coordinates,

$$w_i = a_{ij} w_j + c_{ij}, \qquad g_i = a_{ij} g_j + c_{ij}, \qquad |a_{ij}| = 1.$$

The identity $w_i - g_i = a_{ij}(w_j - g_j)$ makes $\rho = |w_i - g_i|^2$ global. Since g_i is antiholomorphic,

$$\rho_{z\bar{z}} = |(g_i)_{\bar{z}}|^2, \qquad \rho_{z\bar{w}_i} = \rho_{w_i \bar{z}} = 0, \qquad \rho_{w_i \bar{w}_i} = 1.$$

Thus ρ is smooth and psh. On a finite relatively compact trivializing cover of the compact base, the g_i are bounded, so $\rho \le c$ bounds each fiber coordinate. Its sublevels are compact.

For $d \ge 2$, let $B' \to B$ be the finite étale Galois cover associated to the kernel of the permutation monodromy $\pi_1(B) \to \mathfrak{S}_d$ of $C \to B$. Form

$$S' = S \times_B B', \qquad p : S' \to S, \qquad C' = p^{-1}(C) = C_1 \sqcup \cdots \sqcup C_d.$$

The C_i are sections with square zero: their normal bundles are pullbacks of $N_{C/S}$ along finite covers. If $d = 2$, these sections identify the ruling with $\mathbf{P}(\mathcal{O}_{B'} \oplus L)$, where $\deg L = 0$, and $S' \setminus C' = L^\times$. For a unitary flat metric, put $t(v) = \log \|v\|^2$. Then $dd^c t = 0$ on $L^\times$, and

$$\mathcal{L}_{t^2}(\xi) = 2|\partial t(\xi)|^2 \ge 0, \qquad \{t^2 \le c\} = \{e^{-\sqrt{c}} \le \|v\|^2 \le e^{\sqrt{c}}\}$$

is compact. Hence t^2 is a smooth psh exhaustion.

If $d \ge 3$, the unique fiberwise Möbius map sending C_1, C_2, C_3 to $0, 1, \infty$ identifies S' with $B' \times \mathbf{P}^1$. Every remaining section is a holomorphic map $B' \to \mathbf{P}^1 \setminus \{0, 1, \infty\}$; it is constant by the maximum principle. Thus $S' \setminus C' = B' \times (\mathbf{P}^1 \setminus Q)$ for a finite set Q, and an exhaustion of the punctured fiber pulls back to one of the product.

In either case let $\rho' \ge 0$ be the exhaustion and G the deck group. The function

$$\widehat{\rho} = \frac{1}{|G|} \sum_{g \in G} \rho' \circ g$$

is smooth, psh, and G-invariant. Since $\widehat{\rho} \ge \rho'/|G|$, its sublevels are compact. It descends under the finite étale map p to a smooth psh exhaustion of $S \setminus C$. Restrict this function to a collar at E and apply Lemma 5.1(ii) to glue all boundary collars. □

5.4. **Square-zero curves.** For a smooth square-zero curve, infinite Ueda type is a formal condition. A unitary flat neighborhood requires convergent defining functions $w_i = t_{ij} w_j$, $|t_{ij}| = 1$.

Proposition 5.11. *Let $C \subset S$ be smooth, connected, and square-zero, with S compact Kähler. Put $N = N_{C/S}$, let $\chi : H_1(C, \mathbb{Z}) \to \mathrm{U}(1)$ be its unitary character, and set $K = \ker[H_1(C, \mathbb{Z}) \to H_1(S, \mathbb{Z})]$. The complement $S \setminus C$ has a smooth psh exhaustion under any of the following conditions:*

$$C \text{ has finite Ueda type or a unitary flat neighborhood};$$

$$N \text{ is torsion or Diophantine}; \qquad \chi(K) \text{ is finite};$$

$$q(S) \ge g(C); \qquad g(C) = 1.$$

Here Diophantine means $\operatorname{dist}(N^m, \mathcal{O}_C) \ge c m^{-\tau}$ for some $c, \tau > 0$ and all $m \ge 1$. Only the elliptic-curve condition uses [CFS+26, Theorem 1.6].

Proof. Finite type gives a smooth strictly psh collar [CLPT18, Theorem 2.5]. In a unitary flat neighborhood, $w_i = t_{ij}w_j$ with $|t_{ij}| = 1$, so the functions $-\log|w_i|^2$ agree and form a pluriharmonic divergent collar. In infinite type, Ueda's convergence theorem [Ued83, Theorem 3] supplies such a neighborhood when N is torsion or Diophantine.

Suppose $\chi(K)$ is finite, and let k annihilate this finite group. Write $j_* : H_1(C,\mathbb{Z}) \to H_1(S,\mathbb{Z})$ for the inclusion map. The formula

$$\chi_0(j_*\gamma) = \chi(\gamma)^k$$

defines a character on $\operatorname{im} j_*$, because two lifts differ by an element of K. Divisibility of $\mathrm{U}(1)$ extends it to a character $\widetilde{\chi} : H_1(S,\mathbb{Z}) \to \mathrm{U}(1)$, with $\widetilde{\chi} \circ j_* = \chi^k$. For non-torsion N of infinite type, [CLPT18, Theorem 4.4] then gives an effective divisor E disjoint from C with $[E] = m[C]$ for some $m > 0$. The line bundle $\mathcal{O}_S(mC - E)$ has zero real Chern class and hence a unitary flat metric on the compact Kähler surface. Its canonical meromorphic section $\sigma = s_C^m / s_E$ satisfies

$$\operatorname{div}(\sigma) = mC - E, \qquad dd^c \log\|\sigma\|^2 = m[C] - [E].$$

On a neighborhood of C disjoint from E, the function $-m^{-1}\log\|\sigma\|^2$ is therefore pluriharmonic off C and equals $-\log|w|^2$ plus a smooth function. It is the required divergent collar.

For $q(S) \geq g(C)$, if $r : H^1(S,\mathcal{O}_S) \to H^1(C,\mathcal{O}_C)$ is not injective, [CLPT18, Proposition 2.7] makes a multiple of C a fiber, and the punctured-base construction applies. Otherwise r is an isomorphism: its domain has dimension $q(S) \geq g(C)$, and injectivity forces equality. Hodge decomposition and complex conjugation then give an isomorphism $H^1(S,\mathbb{Q}) \to H^1(C,\mathbb{Q})$, whose dual is $j_* \otimes \mathbb{Q}$. Consequently $K \otimes \mathbb{Q} = 0$. As a subgroup of the free abelian group $H_1(C,\mathbb{Z})$, K is torsion-free, hence $K = 0$, and the preceding character argument applies. Finally, if $g(C) = 1$, [CFS+26, Theorem 1.6] rules out infinite type without a unitary flat neighborhood. Apply Lemma 5.1(ii) in all cases. □

6. Holomorphic convexity for curve fiberings

Complex spaces in this section are reduced, Hausdorff, and countable at infinity; smooth functions are understood in local embeddings. Fiber components refer to reduced supports.

Theorem 6.1. *Let X have pure dimension $n \geq 2$ and admit a continuous real-valued psh exhaustion ρ. If a holomorphic map*

$$F : X \longrightarrow \mathbb{C}^{n-1} \quad \textit{satisfies} \quad \dim F^{-1}(z) \leq 1 \qquad (z \in F(X)),$$

then X is holomorphically convex and admits a smooth psh exhaustion.

For smooth X, the convexity argument follows [Vîj22b, Corollary 4.2 and pp. 463–464]. For singular X, fiber cycles and a uniform area bound give the required stability.

6.1. **Compact fiber components.** Write $F = (f_1, \dots, f_{n-1})$. On each local irreducible branch of X at x, the fiber through x is defined by the $n-1$ equations $f_k = f_k(x)$. The analytic dimension theorem therefore gives

$$\dim_x F^{-1}(F(x)) \geq n - (n-1) = 1.$$

Together with the fiber bound, this proves that every nonempty fiber is pure one-dimensional, also after restriction to a local branch. Consider the graph and its projection

$$\iota : X \longrightarrow \Gamma_F \subset \mathbb{C}^{n-1} \times X, \qquad x \longmapsto (F(x), x), \qquad p : \Gamma_F \longrightarrow \mathbb{C}^{n-1}.$$

The map ι is an isomorphism and $p \circ \iota = F$. The reduced graph, with multiplicity one on its components, defines an analytic family of geometric fiber cycles over the normal base [BM98, Chapter I, §1(b), p. 816]:

$$\mathcal{F}_z = \sum_j m_j(z)[C_j(z)], \qquad m_j(z) \in \mathbb{Z}_{>0}, \qquad |\mathcal{F}_z| = F^{-1}(z). \tag{6.1}$$

Empty fibers correspond to the zero cycle. These cycles are locally finite; their components need not be compact.

Lemma 6.2 (Uniform area)**.** *Fix a Hermitian metric h on X. Given compact $L \subset X$ and $z_0 \in \mathbb{C}^{n-1}$, there are a neighborhood B of z_0 and $A > 0$ such that*

$$\int_C h \le A$$

for every compact irreducible component $C \subset L$ of a fiber over B.

Proof. Choose $\chi \in C_c^\infty(X)$ with $0 \le \chi \le 1$ and $\chi = 1$ near L. The map

$$I : \mathbb{C}^{n-1} \longrightarrow \mathbb{R}_{\ge 0}, \qquad I(z) = \int_{\mathcal{F}_z} \chi h$$

is continuous by integration of compactly supported continuous forms on an analytic family of cycles [BM14, Chapter IV, Proposition 2.3.1]. The integral is finite because χ has compact support. Choose B so that $I(z) < I(z_0) + 1$ on B and set $A = I(z_0) + 1$. If $C = C_j(z) \subset L$, its multiplicity in (6.1) is at least one, and positivity gives

$$\int_C h = \int_C \chi h \le m_j(z) \int_C \chi h \le \sum_k m_k(z) \int_{C_k(z)} \chi h = I(z) \le A.$$

□

Lemma 6.3 (Stability)**.** *Let $K \subset X$ be compact. If every irreducible component of $F^{-1}(z_0)$ meeting K is noncompact, the same holds for all fibers over a neighborhood of z_0.*

Proof. Suppose the conclusion fails. There are $z_j \to z_0$, compact irreducible components $C_j \subset F^{-1}(z_j)$, and $x_j \in C_j \cap K$. After taking a subsequence, $x_j \to x \in K$. Let $\nu_j : \widetilde{C}_j \to C_j$ be the normalization. The function $\rho \circ \nu_j$ is subharmonic on the compact connected Riemann surface $\widetilde{C}_j$, so it is constant. Thus, with $R = \max_K \rho$,

$$\rho|_{C_j} = \rho(x_j) \le R, \qquad C_j \subset L := \{\rho \le R\} \Subset X.$$

Lemma 6.2 bounds $\int_{C_j} h$ for all sufficiently large j. Bishop compactness for cycles on complex spaces [Bar99, Theorem 1] yields a convergent subsequence

$$[C_j] \longrightarrow T \quad \text{in the space of compact effective one-cycles of } X.$$

First, $x \in |T|$: otherwise choose an open neighborhood W of x with $\overline{W} \cap |T| = \varnothing$. Upper semicontinuity of compact cycle supports would give $C_j \subset X \setminus \overline{W}$ for large j, whereas $x_j \in W$ for large j. Next, local convergence of integration currents gives $|T| \subset L$, since every test form supported in $X \setminus L$ has zero integral on every C_j. If $y \notin F^{-1}(z_0)$, choose a neighborhood W_y such that $z_j \notin F(W_y)$ for large j. Then $C_j \cap W_y = \varnothing$, and the same current convergence gives $|T| \cap W_y = \varnothing$. Consequently,

$$x \in |T| \subset L \cap F^{-1}(z_0).$$

An irreducible component of $|T|$ through x is a compact analytic curve in the pure one-dimensional fiber $F^{-1}(z_0)$. It is a full irreducible component of that fiber: a proper analytic subset of an irreducible curve has dimension zero. This contradicts the hypothesis. □

6.2. **Convexity and reduction.** We use three complex-space criteria from [Vîj22b, Propositions 2.2, 2.4 and Corollary 2.5]:

(i) A complex space with a continuous psh function u is holomorphically convex if and only if every $\{u < t\}$ is.
(ii) A pseudoconvex space locally holomorphically convex over a Stein base is holomorphically convex.
(iii) If $\pi : M \to S$ is holomorphic, S is Stein, and $D \subset M$ is pseudoconvex, then D is holomorphically convex provided $\overline{D}^M \to S$ is proper and a neighborhood of $\partial_M D$ has Stein reduced fibers.

Here pseudoconvexity means the existence of a continuous psh exhaustion, and local holomorphic convexity over S means that every point of S has a neighborhood V whose inverse image is holomorphically convex.

For criterion (iii), we record the exhaustion step in the cited proof. After shrinking a base neighborhood $V \Subset S$, its finite Stein-neighborhood construction gives smooth functions h_j and nonnegative base cutoffs τ_j for which

$$h_0 = \sum_j (\tau_j \circ \pi) h_j, \qquad h = h_0 + \Lambda \phi \circ \pi$$

is strictly psh on a collar W of the boundary over $\overline{V}$; here ϕ is strictly psh on the base. The product rule and Cauchy–Schwarz on the finitely many local embeddings give constants $a, b, c > 0$ such that

$$\mathcal{L}_{h_0}(\xi) \geq a\|\xi\|^2 - b\|d\pi(\xi)\|^2, \qquad \mathcal{L}_{\phi\circ\pi}(\xi) \geq c\|d\pi(\xi)\|^2.$$

Taking $\Lambda c \geq b$ makes h strictly psh on the collar. The functions are smooth on a neighborhood of $\overline{D} \cap \pi^{-1}(\overline{V})$. Let η_D be a continuous psh exhaustion of D and put

$$\psi = \eta_D + h \quad \text{on } D \cap \pi^{-1}(V).$$

The set

$$H = \big(\overline{D} \cap \pi^{-1}(\overline{V})\big) \setminus W$$

is a compact subset of D. Choose a constant A larger than $\max_H \psi$ when H is nonempty. Then $\{\psi > A\} \subset W$, where ψ is strictly psh. For every compact $T \subset V$, the function h has a lower bound m_T on $\overline{D} \cap \pi^{-1}(T)$, and

$$\{\psi \leq t\} \cap \pi^{-1}(T) \subset \{\eta_D \leq t - m_T\} \cap \pi^{-1}(T) \Subset D.$$

The set on the left is closed in the compact set on the right. Thus ψ is a vertical exhaustion and defines a 1-convex map. Theorem 2.3 and Proposition 2.4 of [Vîj22b] give criterion (iii).

Lemma 6.4. *For every $\lambda \in \mathbb{R}$ and $z_0 \in \mathbb{C}^{n-1}$, there are $c > \lambda$ and a ball $B = B_r(z_0)$ such that*

$$D = F^{-1}(B) \cap \{\rho < c\} \quad \textit{and} \quad F^{-1}(B) \cap \{\rho < \lambda\}$$

are holomorphically convex.

Proof. Let S_0 be the set of constant ρ-values on compact irreducible components of $F^{-1}(z_0)$. For each R, every component whose value is at most R lies in the compact set $\{\rho \leq R\}$. The locally finite family of irreducible components of the central fiber has only finitely many members meeting this set. Therefore

$$\#\big(S_0 \cap (-\infty, R]\big) < \infty.$$

Choose $\lambda < a < c < b$ with $[a,b] \cap S_0 = \varnothing$ and put $K = \{a \le \rho \le b\}$. This is compact, and no compact component of $F^{-1}(z_0)$ meets it. Lemma 6.3 gives a ball $B = B_r(z_0)$ such that no compact component of a fiber over B meets K. Set

$$M = F^{-1}(B), \qquad D = M \cap \{\rho < c\}, \qquad U = M \cap \{a < \rho < b\}.$$

Continuity of ρ gives $\partial_M D \subset \{\rho = c\} \cap M \subset U$. If a reduced fiber of $F|_U$ had a compact irreducible component C, then C would be closed and analytic in X: it is analytic in U, and each point of $X \setminus C$ has a neighborhood disjoint from C. Its dimension would force it to be a full component of the corresponding F-fiber. This is impossible because C meets K. Hence every reduced U-fiber is Stein by the criterion for curves without compact irreducible components [Vîj22a, §3, p. 1663].

On D, define

$$\eta = \max\left\{\frac{1}{c-\rho}, \frac{1}{r^2 - \|F - z_0\|^2}\right\}.$$

For $\alpha \in \mathbb{R}$, the function $g_\alpha(t) = (\alpha - t)^{-1}$ on $t < \alpha$ satisfies

$$g'_\alpha(t) = (\alpha - t)^{-2} > 0, \qquad g''_\alpha(t) = 2(\alpha - t)^{-3} > 0.$$

Both terms are continuous and psh, so the same holds for η. For $t > 0$,

$$\{\eta \le t\} = \{\rho \le c - 1/t\} \cap F^{-1}\{\|z - z_0\|^2 \le r^2 - 1/t\} \Subset D;$$

for $t \le 0$ the sublevel is empty. This proves that D is pseudoconvex. Furthermore, for compact $T \subset B$,

$$\overline{D}^{\,M} \cap F^{-1}(T) \subset \{\rho \le c\} \cap F^{-1}(T).$$

The left side is closed in the compact set on the right, so $\overline{D}^{\,M} \to B$ is proper. Criterion (iii) now gives holomorphic convexity of D. Since $\lambda < c$,

$$F^{-1}(B) \cap \{\rho < \lambda\} = \{x \in D : \rho(x) < \lambda\},$$

and criterion (i) gives the second assertion. □

Proof of Theorem 6.1. For $X(\lambda) = \{\rho < \lambda\}$, Lemma 6.4 makes $F|_{X(\lambda)} : X(\lambda) \to \mathbb{C}^{n-1}$ locally holomorphically convex. The continuous psh function $\eta_\lambda = (\lambda - \rho)^{-1}$ exhausts $X(\lambda)$, since

$$\{\eta_\lambda \le t\} = \{\rho \le \lambda - 1/t\} \Subset X(\lambda) \qquad (t > 0).$$

Criterion (ii) makes every $X(\lambda)$ holomorphically convex, and criterion (i) then applies to ρ on X. Let $q : X \to Z$ be the proper Remmert reduction onto its Stein space Z, and let u be a smooth strictly psh exhaustion of Z. Holomorphic pullback preserves smoothness in local embeddings and plurisubharmonicity, while properness gives

$$u \circ q \in C^\infty(X) \cap \mathrm{PSH}(X), \qquad \{u \circ q \le t\} = q^{-1}(\{u \le t\}) \Subset X.$$ □

Corollary 6.5 (Simultaneous contraction)**.** *The Remmert reduction $q : X \to Z$ has connected fibers, $q_*\mathcal{O}_X = \mathcal{O}_Z$, and $F = \widetilde{F} \circ q$ for a holomorphic $\widetilde{F} : Z \to \mathbb{C}^{n-1}$. Its nontrivial fibers are exactly the maximal connected unions of compact irreducible fiber curves of F.*

Proof. The sheaf identity identifies $\mathcal{O}(Z)$ with $\mathcal{O}(X)$ by pullback. For each coordinate f_k of F, let $\widetilde{f}_k \in \mathcal{O}(Z)$ be its unique preimage. Then

$$\widetilde{F} = (\widetilde{f}_1, \dots, \widetilde{f}_{n-1}), \qquad f_k = \widetilde{f}_k \circ q, \qquad F = \widetilde{F} \circ q.$$

Every q-fiber is therefore a compact connected analytic subset of an F-fiber. If it is not a point, the dimension bound and local finiteness make it a finite connected union of compact irreducible components of that F-fiber.

Conversely, q contracts each compact irreducible curve: its image is a compact analytic subset of the Stein space Z, hence is finite and, by connectedness, a point. Consider a connected union E of compact irreducible components of one F-fiber. The normalization maximum principle makes ρ constant on each component. Intersecting components have the same value; local finiteness and connectedness connect all components by finite chains of intersections. Thus $\rho|_E = a$ for one $a \in \mathbb{R}$, so $E \subset \{\rho \leq a\}$. Only finitely many fiber components meet this compact set. Hence E is a finite connected union and q is constant on E. It follows that the maximal such unions are precisely the nontrivial q-fibers. □

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